HaynesPro®

Data Exchange Web Services V3
VESA

- revision 24.4.1-WS0 -

Table of Contents

1 Identification

1.1 Structure

Identification means the identification of a car. In order to display information about a car, it is necessary to identify the exact type first.

Identification is based on objects with this structure:

```
ExtCarType {
	Integer		order;
	Integer		id;
	String		name;
	String		remark;
	String		fullName;
	Integer		level;
	String		madeFrom;
	String		madeUntil;
	String		engineCode;
	Integer		capacity;
	Integer		output;
	String		modelPictureMimeDataName;
	Integer		numberId;
	String		publisherId;
	Integer		superCarTypeId;
	String		fuelType;
	String[]	subjects;
	HashMap		subjectsByGroup;
}
```

This object can contain information about a MAKE (like ALFA ROMEO, AUDI, VOLKSWAGEN ...), a MODEL (like Golf IV is for VOLKSWAGEN) and a TYPE (like the 1.4 16V AKQ is for Golf IV):

SELECT A MAKE:

ALFA ROMEO	FORD	MAZDA	SAAB
AUDI	HONDA	MERCEDES-BENZ	SEAT
BMW	HYUNDAI	MG	SKODA
BUICK	IVECO	MINI	SMART
CADILLAC	JAGUAR	MITSUBISHI	SSANGYONG
CHEVROLET (DAEWOO)	JEEP	NISSAN	SUBARU
CHEVROLET (USA)	KIA	OPEL / VAUXHALL	SUZUKI
CHRYSLER	LADA	PEUGEOT	TOYOTA
CITROEN	LANCIA	PORSCHE	TVR
DACIA	LAND & RANGE ROVER	RENAULT	VOLKSWAGEN
DAIHATSU	LEXUS	ROVER	VOLVO
FIAT	LINCOLN		

Figure 3.1.1 – Make selection

VOLKSWAGEN (« Return to Makes selection)

SELECT A MODEL: Filter: Reset

Make, Model	Years build
» VOLKSWAGEN Eos	2006 - ...
» VOLKSWAGEN Fox	2004 - ...
» VOLKSWAGEN Golf II	1983 - 1992
» VOLKSWAGEN Golf III (Cabrio 1998->)	1991 - 1998
» VOLKSWAGEN Golf IV	1998 - 2003
» VOLKSWAGEN Golf V	2003 - ...
» VOLKSWAGEN Golf Plus	2005 - ...
» VOLKSWAGEN Jetta II	1984 - 1992
» VOLKSWAGEN Jetta	2005 - ...
» VOLKSWAGEN LT	1983 - 1996
» VOLKSWAGEN LT	1996 - 2006
» VOLKSWAGEN Lupo	1998 - 2005

Figure 3.1.2 – Model selection

SELECT A TYPE: Filter: Reset

Model, Type	Years build	Engine code	Capacity	Output
» Golf IV 1.4 16V	1997 - 2000	AKQ	1390cc	55kW
» Golf IV 1.4 16V	1998 - 2003	AHW	1390cc	55kW
» Golf IV 1.4 16V	1999 - 2000	APE	1390cc	55kW
» Golf IV 1.4 16V	2000 - 2001	AXP	1390cc	55kW
» Golf IV 1.4 16V	2001 - 2003	BCA	1390cc	55kW
» Golf IV 1.6	1997 - 2003	AKL	1595cc	74kW
» Golf IV 1.6	1998 - 2003	AEH	1595cc	74kW
» Golf IV 1.6	1998 - 2003	APF	1595cc	74kW
» Golf IV 1.6	2000 - 2002	AVU	1595cc	75kW
» Golf IV 1.6	2002 - 2003	BFQ	1595cc	75kW
» Golf IV 1.6 16V	1999 - 2000	ATN	1597cc	77kW
» Golf IV 1.6 16V	2000 - 2000	AUS	1597cc	77kW
» Golf IV 1.6 16V	2000 - 2001	AZD	1597cc	77kW
» Golf IV 1.6 16V	2001 - 2003	BCB	1597cc	77kW
» Golf IV 1.6 FSi	2002 - 2003	BAD	1597cc	81kW
» Golf IV 1.8	1997 - 2000	AGN	1781cc	92kW
» Golf IV 1.8 Turbo	1997 - 2003	AGU	1781cc	110kW
» Golf IV 1.8 Turbo	1998 - 2000	AQA	1781cc	110kW

Figure 3.1.3 -Type selection

This is a general object for all the 3 levels (make, model and type) and there are some elements that are specific to only one of them and elements (for example, the model picture will be filled only for a model) that are used for all of them (like the name).

Next, we will give an explanation for each element in the object:

```
ExtCarType {
      Integer     order;
      Integer     id;
      String      name;
      String      remark;
      String      fullName;
      Integer     level;
      String      madeFrom;
      String      madeUntil;
      String      engineCode;
      Integer     capacity;
      Integer     output;
      String      modelPictureMimeDataName;
      Integer     superCarTypeId;
      String      fuelType;
      String[]    subjects;
      HashMap     subjectsByGroup;
}
```

- **order**: this is a number that has a meaning only when you get more responses. For example, you give a model and you want all the types that the model has. You will get a set types and they will be ordered based on an internal criteria (for example by engine capacity as the primary criteria, the year that they started producing it as a second criteria and the engine code as the third criteria; by internal criteria we mean that this is not explicitly expressed, it cannot be changed by the user and it can change in case we find a better way to order them). Normally, the order in the response XML is the same as the order number, but in case there is a client-application that doesn't preserve the order in which the elements were received.
- **id:** this is a number that uniquely identifies a car-type in this application. Because these ids are generated based on the order in the database, they may be different from a version of the database to the other (so it will be unique within a session of usage, but don't store it and expect to be the same in a few months or weeks). We are in the process of making them stable so they don't change anymore, but until then you can use "numberId" and "publisherId" together in case you need a unique id over a longer time.
- **name**: this is the name of the make, model or type. If the object is a make, it can be "VOLKSWAGEN", if it is a model, it can be "Golf" and if it is a type it can be "1.4 16V" (you will know what kind of object it is based on the "level")
- **remark**: this element, in case it is not null, comes to complete the name. It can be found only in

models. For example, for a Volkswagen Golf, it can be “I”, “II”, “III”, “IV”, “V” or “Plus” (see figure 3.2)

- **fullName**: this property is also the name of the object, but it includes the name of the more general objects too. For a make, it can be “Volkswagen”, for a model it can be “Volkswagen Golf IV” and for a type it can be “Volkswagen Golf IV 1.4 16V”.
- **level**: using this you can identify the type of the object: make, model and types:
  - level = 1 -> the object is a Make
  - level = 2 -> the object is a Model
  - level = 3 -> the object is a Type
- **madeFrom**: this property encodes the year that the model or the type has started to be made. The format is: yyyy-MM (for example 1997-01 means January 1997). This property should never be null for a Model or a Type (there should always be a start year for a model or for a specific type).
- **madeUntil**: this property encodes the year and month until that the model started to be out of production. It is encoded the same: yyyy-MM (exactly as madeFrom). The only difference is that this property can be null.
- **engineCode**: the engine code is available only for Types. For example, for “Volkswagen Golf IV 1.4 16V” you can have engineCode = “AKQ” (see figure 3.1.3)
- **capacity**: this is the capacity of the engine (so it is available only for Types). For “Volkswagen Golf IV 1.4 16V”, it is 1390cc (see figure 3.1.3)
- **output**: this is the output of the engine (so it is available only for Types). For example, for “Volkswagen Golf IV 1.4 16V”, it is 55kW (see figure 3.1.3)
- **modelPictureMimeDataName**: this property contains an URL to an image and it is available only for Models and for Types (but for types it is always the one from their model). For example, for “Volkswagen Golf IV”, it is “http://www.haynespro-services.com:8080/workshop/images/golf_4_dba_soest.jpg”
- **superCarTypeId**: this id is the id of the super-object. For example, if this is part of a Type, it will be the id of the model. If it is part of a model object, it will represent the id of the Make. If it is part of the Make, it will be null.
- **fuelType**: the fuel type can be “PETROL”, “DIESEL” and, very rarely, when it is not known, it will be null (note: this is a constant/enumeration and it will not be translated)
- **subjects**: this property is an array of constant elements. It specifies what kind of information is available for the specific car-type. The values that you can expect to find in this property are:
  - `ADJUSTMENTS`

- MAINTENANCE
- LUBRICANTS
- DRAWINGS
- EOBD_LOCATIONS
- FUSE_LOCATIONS
- ENGINE_LOCATIONS
- MANAGEMENT
- STORIES
- WARNING_LIGHTS
- WIRING_DIAGRAMS
- REPAIR_TIMES
- TIMING_REPAIR_MANUALS
- VESA_ENGINE
- VESA_ABS
- SHOW_VESA
- TSB

In the next table you can see the relations between the operations of the web-service and these subjects:

Operation name	**Subject**
getAdjustments()	ADJUSTMENTS
getMaintenanceSystems()	MAINTENANCE
getLubricants()	LUBRICANTS
getDrawings()	DRAWINGS
getEobdLocations()	EOBD_LOCATIONS
getFuseLocations()	FUSE_LOCATIONS
getEngineLocations()	ENGINE_LOCATIONS
getManagementSystems()	MANAGEMENT
getStoriesOverview()	STORIES
getWiringDiagrams()	WIRING_DIAGRAMS
getRepairtimeTypes()	REPAIR_TIMES
getTimingStoryOverview()	TIMING_REPAIR_MANUALS
getTSBSystems()	TSB
getWarningLights()	WARNING_LIGHTS

Table 3.1.1 – Main subjects

The 3 elements that contain VESA (VESA_ENGINE, VESA_ABS and SHOW_VESA) are related to data that is not provided by this web-service. Only two of them (VESA_ENGINE and VESA_ABS) are are a direct result of existing data, while the third (SHOW_VESA) is calculated based on a formula that includes MANAGEMENT, ABS_ELECTRONICAL, VESA_ENGINE (the new equivalent of the MANAGEMENT) and VESA_ABS (the new equivalent of ABS_ELECTRONICAL). We recommend to use the flash component that we provide (please contact us for it) that shows the VESA data only when "SHOW_VESA" is present in the subjects.

- **subjectsByGroup**: we divide information into 7 main groups. These groups are:
  - ENGINE
  - TRANSMISSION
  - STEERING
  - BRAKES
  - EXTERIOR
  - ELECTRONICS
  - QUICKGUIDES

This property is a map that has as key one of this elements as a string and as value an array of strings with a new set of subjects (the array will be represented in this case as a String of elements separated by comas). See the next table for the relations between these subjects and the operations that the web-service offers:

Group	Subject	Operation
ENGINE	ADJUSTMENTS	getAdjustmentsByGroup(..., "ENGINE")
	LUBRICANTS	getLubricantsByGroup(..., "ENGINE")
	STORIES	getStoryOverviewByGroup(..., "ENGINE")
	DRAWINGS	getDrawingsByGroup(..., "ENGINE")
	MANAGEMENT	getManagementSystems()
	REPAIR_TIMES	getRepairtimeSubnodesByGroup(...,"ENGINE")
	TSB	GetTSBSystems(...,"ENGINE")
TRANSMISSION	ADJUSTMENTS	getAdjustmentsByGroup(..., "TRANSMISSION")
	LUBRICANTS	getLubricantsByGroup(..., "TRANSMISSION")
	STORIES	getStoryOverviewByGroup(..., "TRANSMISSION")
	DRAWINGS	getDrawingsByGroup(..., "TRANSMISSION")
	AUTOMATIC_TRANSMISSION	getManagementSystems()
	REPAIR_TIMES	getRepairtimeSubnodesByGroup(...,"TRANSMISSION")
	TSB	GetTSBSystems(...,"TRANSMISSION")
STEERING	ADJUSTMENTS	getAdjustmentsByGroup(..., "STEERING")
	LUBRICANTS	getLubricantsByGroup(..., "STEERING")

	STORIES	getStoryOverviewByGroup(..., "STEERING")
	DRAWINGS	getDrawingsByGroup(..., "STEERING")
	REPAIR_TIMES	getRepairtimeSubnodesByGroup(...,"STEERING")
	TSB	GetTSBSystems(...,"STEERING")
BRAKES	ADJUSTMENTS	getAdjustmentsByGroup(..., "BRAKES")
	LUBRICANTS	getLubricantsByGroup(..., "BRAKES")
	STORIES	getStoryOverviewByGroup(..., "BRAKES")
	DRAWINGS	getDrawingsByGroup(..., "BRAKES")
	ABS_ELECTRONICAL	getManagementSystems()
	REPAIR_TIMES	getRepairtimeSubnodesByGroup(...,"BRAKES")
	TSB	GetTSBSystems(...,"STEERING")
EXTERIOR	ADJUSTMENTS	getAdjustmentsByGroup(..., "EXTERIOR")
	LUBRICANTS	getLubricantsByGroup(..., "EXTERIOR")
	STORIES	getStoryOverviewByGroup(..., "EXTERIOR")
	DRAWINGS	getDrawingsByGroup(..., "EXTERIOR")
	ABS_ELECTRONICAL	getManagementSystems()
	REPAIR_TIMES	getRepairtimeSubnodesByGroup(...,"EXTERIOR")
	AIRCO_WIRING_DIAGRAMS	GetWiringDiagramsByGroup(...,"EXTERIOR")
	TSB	GetTSBSystems(...,"EXTERIOR")
ELECTRONICS	MANAGEMENT	getManagementSystems()
	ABS_ELECTRONICAL	getManagementSystems()
	AUTOMATIC_TRANSMISSION	getManagementSystems()
	WIRING_DIAGRAMS	getWiringDiagramsByGroup(...,"ELECTRONICS")
	FUSES	getFuseLocations()
	TSB	GetTSBSystems(...,"ELECTRONICS")
	WARNING_LIGHTS	getWarningLights()
QUICKGUIDES	ADJUSTMENTS	getAdjustmentsByGroup(..., "QUICKGUIDES")
	LUBRICANTS	getLubricantsByGroup(..., "QUICKGUIDES")
	STORIES	getStoryOverviewByGroup(..., "QUICKGUIDES")
	DRAWINGS	getDrawingsByGroup(..., "QUICKGUIDES")
	TSB	GetTSBSystems(...,"QUICKGUIDES")

Table 3.1.2 – Subjects by groups

Next, we will show, based on a real example, where each element of this structure is used in Workshop ATI Online. For this example, we will use "Volkswagen Golf IV 1.4 16V AKQ" (the same one that is selected in image 3.1.3.

The structure returned by the web-service is (it can be returned different operations, but for this example we suggest calling the “findCarTypes” operation with the parameters “en” for language and “Volkswagen Golf IV AKQ” as search string):

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <findCarTypesResponse xmlns="http://data.webservice.workshop.vivid.nl">
         <findCarTypesReturn>
            <capacity>1390</capacity>
            <engineCode>AKQ</engineCode>
            <fuelType>PETROL</fuelType>
            <fullName>VOLKSWAGEN Golf IV 1.4 16V</fullName>
            <id>26650</id>
               <level>3</level>
               <madeFrom>1997-01</madeFrom>
               <madeUntil>2000-12</madeUntil>
               <modelPictureMimeDataName>http://www.haynespro-services.com:8080/workshop/images/golf_4_dba_soest.jpg</modelPictureMimeDataName>
               <name>1.4 16V</name>
               <order>0</order>
               <output>55</output>
               <remark xsi:nil="true"/>
               <status xsi:nil="true"/>
               <subjects>
                  <item>ADJUSTMENTS</item>
                  <item>MAINTENANCE</item>
                  <item>LUBRICANTS</item>
                  <item>DRAWINGS</item>
                  <item>EOBD_LOCATIONS</item>
                  <item>FUSE_LOCATIONS</item>
                  <item>ENGINE_LOCATIONS</item>
                  <item>MANAGEMENT</item>
                  <item>STORIES</item>
                  <item>WIRING_DIAGRAMS</item>
                  <item>REPAIR_TIMES</item>
               </subjects>
               <subjectsByGroup>
                  <item xmlns:ns1="http://xml.apache.org/xml-soap" xmlns="">
                     <key xsi:type="xsd:string">ELECTRONICS</key>
                     <value xsi:type="xsd:string">FUSE_LOCATIONS,MANAGEMENT,AUTOMATIC_TRANSMISSION,ABS_ELECTRONICAL,WIRING_DIAGRAMS</value>
                  </item>
                  <item xmlns="">
                     <key xsi:type="xsd:string">EXTERIOR</key>
                     <value xsi:type="xsd:string">ADJUSTMENTS,STORIES,DRAWINGS,AIRCO_WIRING_DIAGRAMS,REPAIR_TIMES</value>
                  </item>
```

```
                    <item xmlns="">
                       <key xsi:type="xsd:string">QUICKGUIDES</key>
                       <value xsi:type="xsd:string">ADJUSTMENTS,LUBRICANTS,STORIES,DRAWINGS,REPAIR_TIMES</value>
                    </item>
                    <item xmlns="">
                       <key xsi:type="xsd:string">BRAKES</key>
                       <value xsi:type="xsd:string">ADJUSTMENTS,LUBRICANTS,DRAWINGS,ABS_ELECTRONICAL,REPAIR_TIMES</value>
                    </item>
                    <item xmlns="">
                       <key xsi:type="xsd:string">ENGINE</key>
                       <value xsi:type="xsd:string">ADJUSTMENTS,LUBRICANTS,STORIES,DRAWINGS,MANAGEMENT,REPAIR_TIMES</value>
                    </item>
                    <item xmlns="">
                       <key xsi:type="xsd:string">TRANSMISSION</key>
                       <value xsi:type="xsd:string">ADJUSTMENTS,LUBRICANTS,DRAWINGS,AUTOMATIC_TRANSMISSION,REPAIR_TIMES</value>
                    </item>
                    <item xmlns="">
                       <key xsi:type="xsd:string">STEERING</key>
                       <value xsi:type="xsd:string">ADJUSTMENTS,LUBRICANTS,DRAWINGS,REPAIR_TIMES</value>
                    </item>
                 </subjectsByGroup>
                 <superCarTypeId>7250</superCarTypeId>
              </findCarTypesReturn>
           </findCarTypesResponse>
        </soapenv:Body>
     </soapenv:Envelope>
```

HaynesPro

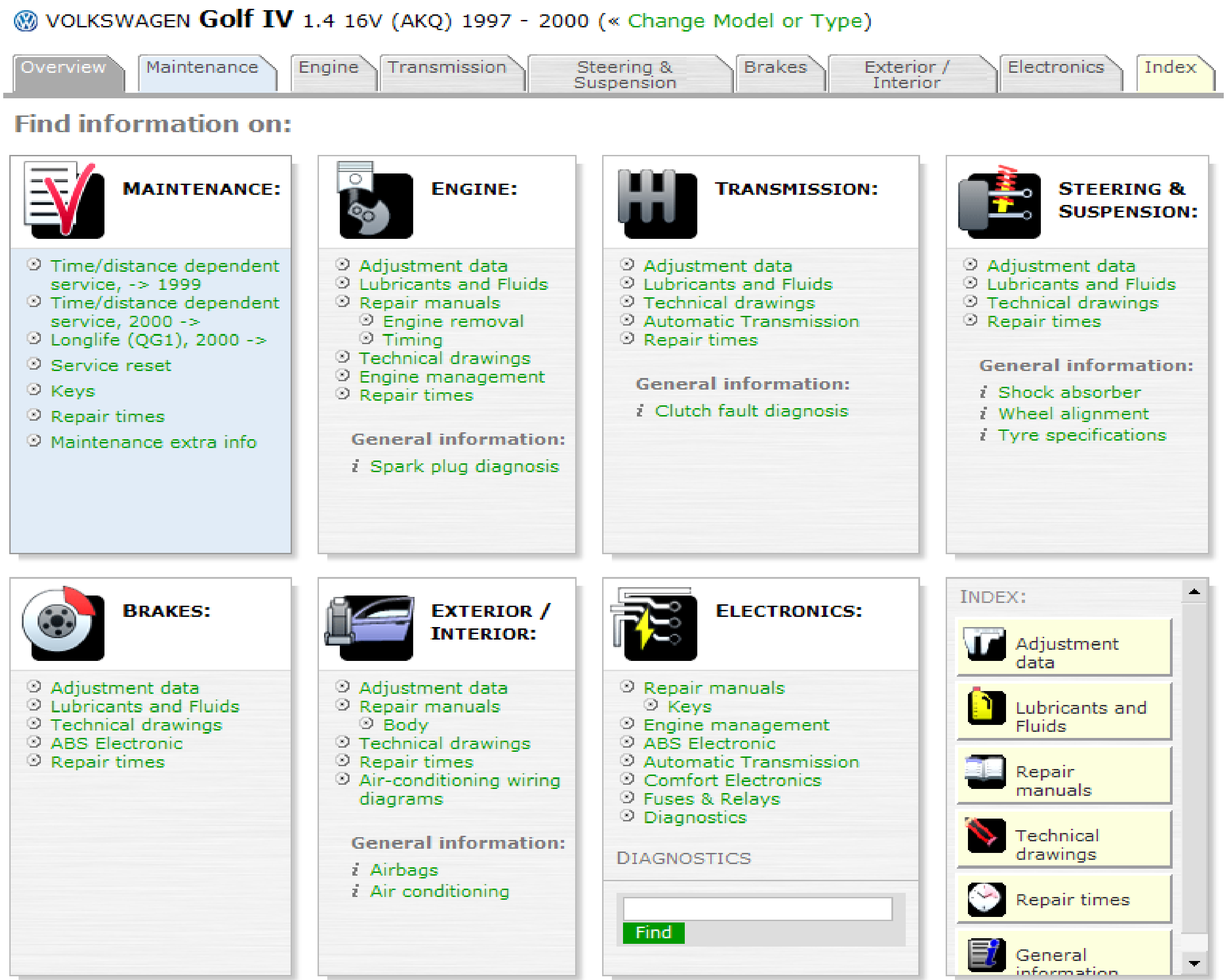


Figure 3.1.4 – Car Home for Volkswagen Golf IV 1.4 16V (AKQ)

The header:

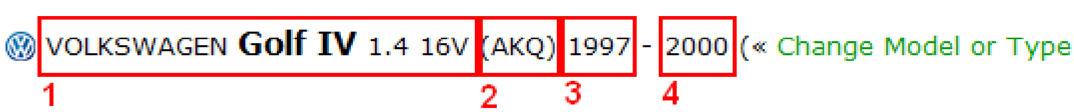


Figure 3.1.5 - Header

```
1 - <fullName>VOLKSWAGEN Golf IV 1.4 16V</fullName>    (line 8)
2 - <engineCode>AKQ</engineCode>                       (line 6)
3 - <madeFrom>1997-01</madeFrom>                       (line 11)
4 - <madeUntil>2000-12</madeUntil>                     (line 12)
```

HaynesPro

Maintenance panel:

Figure 3.1.6 – Maintenance panel

```
1 - <subjects> ... <item>MAINTENANCE</item> ... </subjects>        (line 21)
2 - <subjects> ... <item>REPAIR_TIMES</item> ... </subjects>       (line 30)
```

Engine group panel:

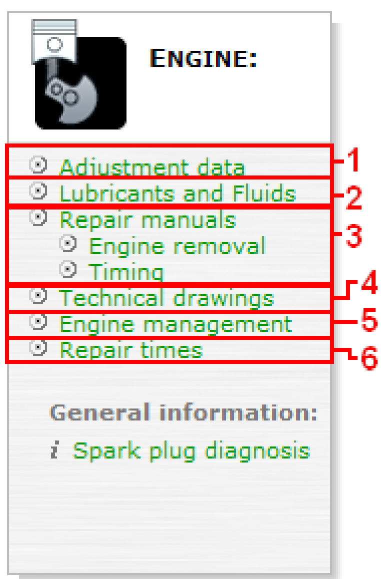


Figure 3.1.6 – Engine group panel

```
<item xmlns="">
      <key xsi:type="xsd:string">ENGINE</key>
```

```
        <value xsi:type="xsd:string">
              ADJUSTMENTS,                    -1
              LUBRICANTS,                     -2
              STORIES,                        -3
              DRAWINGS,                       -4
              MANAGEMENT,                     -5
              REPAIR_TIMES</value>            -6
  </item>
```

Note: You can see that at point 3 (Repair manuals), there are details about how the repair manuals are named. You can get this information by calling one of the operations. They will be detailed in the next chapters (getStoryOverviewByGroup() in this case).

For the rest of the groups (Transmission, Steering, Brakes, Exterior, Electronics and Quickquides), the relation is the same as for Engine (if you check "Engine group panel" and the 3.1.2 table, you can get the relations between Workshop ATI Online and the web-service for the car-type object structure.

2 Vehicle Electronics Smart Assistant (VESA)

This is a complex module providing data related to electronics schemes, components and diagnosis.

2.1 Systems

2.1.1 Structure:

```
SystemGroupCategoryArrayContainer {
      SystemGroupCategory[]    systemGroupCategoryArray;
}
```

```
SystemGroupCategory {
      String              mainGroup;
      Integer             system_group_id;
      Integer             system_group_number;
      String              text;
      System[]            systems;
      SystemGroup[]       systemGroups;
}
```

- **mainGroup**: the group of the system group category. It can be ENGINE, BRAKES, STEERING, TRANSMISSION, EXTERIOR
- **system_group_id:** an integer number representing an internal identifier for the group
- **system_group_number**: this number represents the sort order when there are more such objects in the result
- **text**: a text containing the description of the category and it is language dependent (for example: “Engine Management”, “ABS”, “ABS and Stability”)
- **systems**: this is a list of complex objects representing systems; their structure will be defined next
- **systemGroups**: this is a list of complex objects; sometimes the systems are grouped and the top object represents the group (“ABS and Stability”), and this property contains a list of systems (“ABS” being one and “Stability” being another one)

```
System {
      int          system_type_id;
      String       text;
}
```

- **system_type_id**: an integer number representing an internal identifier for the system type
- **text**: this is a text containing the description of the system and it is language dependent (for example: “Teves MK 20”, “Teves MK 20, -> 10/2001”, “Magneti Marelli 4AV, 05/1999 ->”)

```
SystemGroup {
```

```
    Integer             system_group_id;
    String              text;
    System[]            systems;
    Integer             sg_number;
}
```

- **system_group_id**: an integer number representing an internal identifier for the group
- **systems**: this is a list of complex objects representing systems; their structure will be defined above
- **text**: a text containing the description of the category and it is language dependent (for example: “ABS”)
- **sg_number**: this represents the sort-order

  Note! The SystemGroup object has exactly the same structure as SystemGroupCategory, except for the “systemGroups” property and the name for the sort-order. You can see it as a tree structure where the SystemGroupCategory is the top level and SystemGroup is the second level.

2.1.2 Operation: getSystemsV2

The operation requires the car id and the language name, and returns the data with the structure that is described above.

- car_type – int (this number is the “id” of the Type found in ExtCarType; you can check chapter 3 for more information about this object)
- language - String (this should be a 2 character string, established by the 639-1 ISO; for example, for English it is “en”, for French it is “fr”)

  Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:getSystemsV2>
         <web:vrid>...</web:vrid>
         <web:typeId>26640</web:typeId>
         <web:language>en</web:language>
      </web:getSystemsV2>
   </soapenv:Body>
</soapenv:Envelope>
```

- response (fragment):

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getSystemsV2Response xmlns="http://webservice.elecdata.vivid.nl">
```

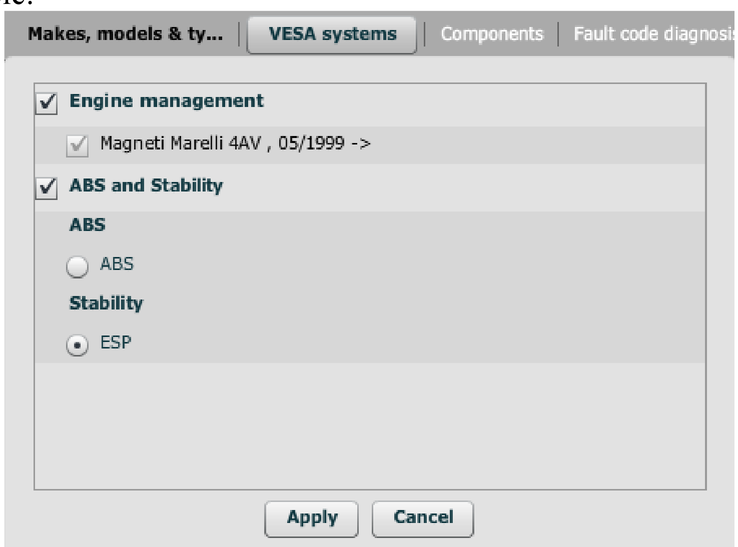
```xml
        <getSystemsV2Return>
           <status xsi:nil="true"/>
           <systemGroupCategoryArray>
              <item>
                 <mainGroup>ENGINE</mainGroup>
                 <systemGroups xsi:nil="true"/>
                    <system_group_id>2</system_group_id>
                    <system_group_number>1</system_group_number>
                    <systems>
                       <item>
                          <system_type_id>2971</system_type_id>
                          <text>Magneti Marelli 4AV, 05/1999 -></text>
                       </item>
                    </systems>
                    <text>Engine management</text>
                 </item>
              </systemGroupCategoryArray>
              <systemGroupCategoryArray>
                 <item>
                    <mainGroup>BRAKES</mainGroup>
                    <systemGroups>
                       <item>
                          <sg_number>0</sg_number>
                          <system_group_id>8</system_group_id>
                          <systems>
                             <item>
                                <system_type_id>106007996</system_type_id>
                                <text>Teves MK 20</text>
                             </item>
                          </systems>
                          <text>ABS</text>
                       </item>
                       <item>
                          <sg_number>1</sg_number>
                          <system_group_id>5</system_group_id>
                          <systems>
                             <item>
                                <system_type_id>106007963</system_type_id>
                                <text>Teves MK 20, 11/2001 -></text>
                             </item>
                             <item>
                                <system_type_id>106006355</system_type_id>
                                <text>Teves MK 20, -> 10/2001</text>
                             </item>
                          </systems>
                          <text>Stability</text>
                       </item>
                    </systemGroups>
                    <system_group_id>12</system_group_id>
                    <system_group_number>2</system_group_number>
                    <systems xsi:nil="true"/>
                    <text>ABS and Stability</text>
                 </item>
```

```
            ...
        </systemGroupCategoryArray>
      </getSystemsV2Return>
    </getSystemsV2Response>
  </soapenv:Body>
</soapenv:Envelope>
```

Note! In the previous example, the first object has only one level and it represents a system-group-category, and the second object has 2 levels, representing a group that has as “systemGroups” a list of system-group-categories.

Each system-group-category has at least one system. The **ids** of the systems (**system_type_id**) together with the **ids** of the system-grup-categories (**system_group_id**) are the elements that are used in the next calls.

Implementation example:



2.2 Networks

2.2.1 Structure

```
NetworkArrayContainer {
	Network[] networkArray;
}
```

- networkArray: this is a list of complex object (Network) which are described next

```
Network {
	int			systemTypeId;
```

```
        Integer[]           systemGroupId;
        String[]            groupName;
        String              text;
        NetworkSignal[]     signals_in;
        NetworkSignal[]     signals_out;
        Integer             sgNumber;
}
```

- **systemTypeId**: this is a number representing the id of the system; it is the same one that is passed as a parameter when requesting the network and it is useful to distinguish between multiple objects (more systemTypes can be requested at the same time and this one makes it possible to identify each one)
- **systemGroupId**: this is a number representing the id of the system-group; it is the same one that is passed as a parameter when requesting the network and it is useful to distinguish between multiple objects
- **groupName**: this is a text containing the description of the group; it is also present in the previous call (of getSystems) (for example: "Engine management")
- **text**: this is a text containing the description of the system; it is also present in the previous call (of getSystem) (for example: "Magneti Marelli 4AV")
- **signals_in**: this is a list of complex objects representing the input signals (the structure of the object is described next)
- **signals_out**: this is a list of complex objects representing the output signals (the structure of the object is described next)
- sgNumber: sort order

Note! The Network object, except for the "signals_in" and "signals_out" properties, doesn't bring any new information compared to the previous call (getSystems). All the values are useful to identify the system when more are selected.

```
NetworkSignal {
        Integer             signal_id;
        String              text;
        SignalInstance[]    instances;
        Integer[]           system_groups;
}
```

- **signal_id**: this is a number that uniquely identifies the signal
- **text**: this is a text that describes the signal (for example: "Camshaft position", "Camshaft speed", "Ignition")
- **instances**: this is a list of complex objects that represent certain components from scheme that are related to this signal; the structure of these complex objects will be described next

- **system_groups**: this is a list of numbers representing the ids of the groups the signal belongs to (see "SystemGroupCategory" and "SystemGroup" from subchapter 4.1.1, Systems – Structure)

```
SignalInstance {
      Integer      component_id;
      String       text;
      String       instance_order;
}
```

- **component_id**: a number representing a unique identifier of the component
- **text**: this is a text representing the description of the component (for example: "Canister purge solenoid 1", "Fuel pump 1", "Injector 2")
- **instance_order**: a number representing the sort order applied on the components for a signal

2.2.2 Operation: "getNetworksV2"

This operation requires as parameters the car id, a list of system ids, a list of group ids and the language.

These are the request parameters and their meaning:

- system_types_ids – a list of numbers, the ids for the system-types
- system_types_groups – a list of numbers, the ids for system-type-groups (the list must have the same length as system_types_ids)
- car_type – int (this number is the "id" of the Type found in ExtCarType; you can check chapter 3 for more information about this object)
- language - String (this should be a 2 character string, established by the 639-1 ISO; for example, for English it is "en", for French it is "fr")

Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
 <soapenv:Header/>
   <soapenv:Body>
      <web:getNetworks>
         <web:vrid>?</web:vrid>
         <!--1 or more repetitions:-->
         <web:system_types_ids>2971</web:system_types_ids>
         <!--1 or more repetitions:-->
         <web:system_types_groups>2</web:system_types_groups>
```

```
            <web:car_type>26640</web:car_type>
            <web:language>en</web:language>
         </web:getNetworks>
      </soapenv:Body>
   </soapenv:Envelope>
```

- response (fragment):

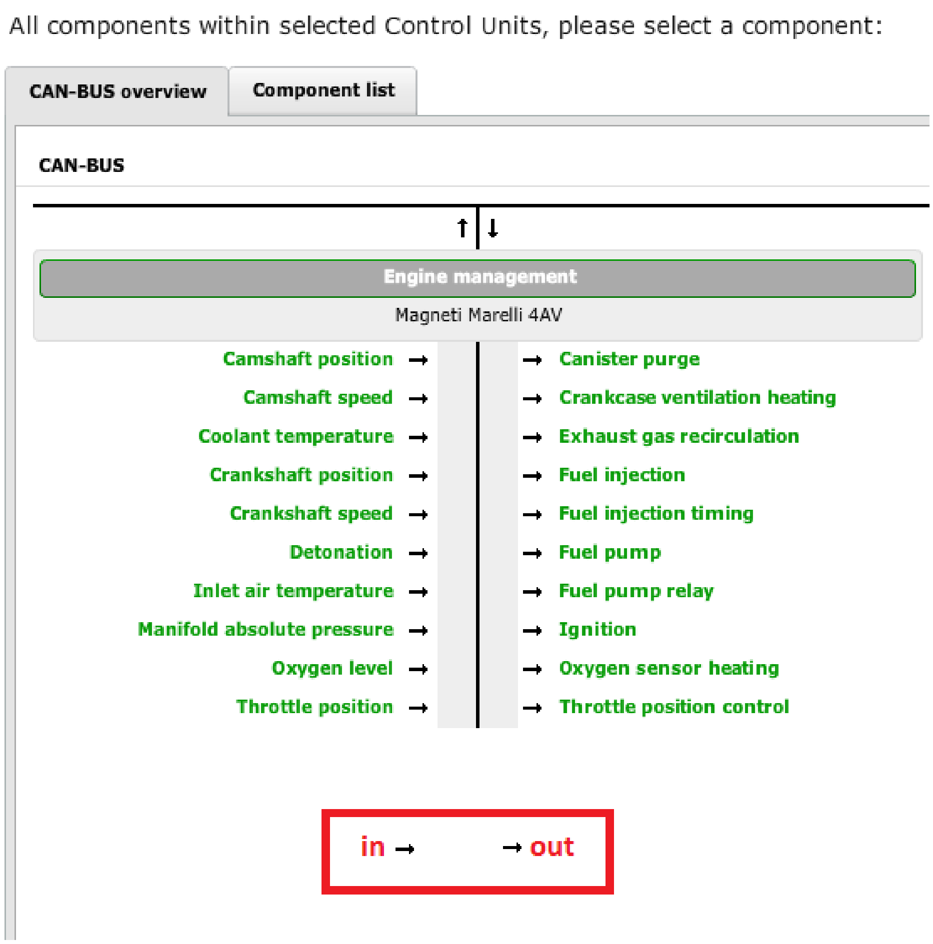
```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getNetworksResponse xmlns="http://webservice.elecdata.vivid.nl">
         <getNetworksReturn>
            <networkArray>
               <item>
                  <groupName>
                     <item>Engine management</item>
                  </groupName>
                     <sgNumber>1</sgNumber>
                     <signals_in>
                        <item>
                           <instances>
                              <item>
                                 <component_id>106231881</component_id>
                                 <instance_order>1</instance_order>
                                 <text>Hall/MRE sensor on the camshaft 1</text>
                              </item>
                           </instances>
                           <signal_id>56</signal_id>
                           <system_groups>
                              <item>2</item>
                           </system_groups>
                           <text>Camshaft position</text>
                        </item>
                        ...
                        <item>
                           <instances>
                              <item>
                                 <component_id>106231929</component_id>
                                 <instance_order>1</instance_order>
                                 <text>Magnetic pick-up on the crankshaft
1</text>
                              </item>
                           </instances>
                           <signal_id>15</signal_id>
                           <system_groups>
                              <item>7</item>
                              <item>9</item>
                              <item>5</item>
                              <item>12</item>
                              <item>2</item>
```

```
            </system_groups>
            <text>Crankshaft speed</text>
         </item>
         ...
      </signals_in>
      <signals_out>
         <item>
            ...
            <signal_id>64</signal_id>
            <system_groups>
               <item>2</item>
            </system_groups>
            <text>Canister purge</text>
         </item>
         ...
         <item>
            <instances>
               <item>
                  <component_id>106231920</component_id>
                  <instance_order>1</instance_order>
                  <text>Injector 1</text>
               </item>
               <item>
                  <component_id>106231913</component_id>
                  <instance_order>2</instance_order>
                  <text>Injector 2</text>
               </item>
               <item>
                  <component_id>106231890</component_id>
                  <instance_order>3</instance_order>
                  <text>Injector 3</text>
               </item>
               <item>
                  <component_id>106231919</component_id>
                  <instance_order>4</instance_order>
                  <text>Injector 4</text>
               </item>
            </instances>
            <signal_id>25</signal_id>
            <system_groups>
               <item>2</item>
            </system_groups>
            <text>Fuel injection</text>
         </item>
         ...
      </signals_out>
      <systemGroupId>
         <item>2</item>
      </systemGroupId>
      <systemTypeId>2971</systemTypeId>
      <text>Magneti Marelli 4AV</text>
   </item>
</networkArray>
```

```
                <status xsi:nil="true"/>
             </getNetworksReturn>
          </getNetworksResponse>
       </soapenv:Body>
    </soapenv:Envelope>
```

Note! The “component_id” and the “systemTypeId” are the two elements that are used from this response for the next calls.

Implementation Example:



2.3 Components

2.3.1 Structure

```
CompleteComponentContainerV10 {
      CompleteComponentV10 completeComponentV10;
      ExtStatus            status;
}


CompleteComponentV10{
      PinsArrayV2        pinTable;
      PinsArrayV2        pinTableExtended;
      ExtSmartLink[]     smartLinks;
      WizardV3[]         diagnosisSteps;
      String             ecuSideImage;
      String             connectorSideImage;
      Boolean            renderable;
      Boolean            hasLocationSystems;
      Integer            componentImageId;
      GenericPart[]      genericParts;
      Boolean            renderable;
      LinkedStoriesV2[]  linkedStories;
      String             localWiringImageConcise;
      String             localWiringImageExtended;
      String             componentDescription;
      String             componentCode;
      String             symbolInstance;

}
```

- **pinTable**: this is a complex object representing a list of components and their pins (can be represented as a table); the structure of this object is presented next
- **pinTableExtended**: this is a complex object representing a list of components and their pins from the extended diagram (can be represented as a table); the structure of this object is presented next
- **smartLinks**: this is a true/false value. It shows links from vesa components to other subjects (currently only to repair manuals) See chapter "6.2.3 Smart Links" in general documentation for more details about Smart Links.

- **diagnosisSteps**: this is a list of complex object representing the steps of the diagnosis for the selected component and details about the measurement instruments involved it he diagnosis
- **localWiringImageConcise**: this is a text that contains a URL to a scheme of the component together with the other important components that it is linked to. The link returns a SVG file (or a JPG if the parameter inside the URL is changed from svg to jpg).
- **localWiringImageExtended**: this is the same as the "localWiringImageConcise", but the URL links to a diagram that could contain more elements (that are on the path to the ground); parameter policy=full has been introduced to load the extended diagram
- **ecuSideImage**: this is a text that may contain a URL to an image representing the ECU

component from a side

- **connectorSideImage**: this is a text that may contain a URL to an image representing the ECU connector from a side
- **renderable**: this is a boolean value that contains “true” if the re is a scheme (subnetXml is not null) and false if there isn't a scheme (subnetXml is null or empty, in which case the ECU-images are not null)
- **hasLocationSystems**: is a boolean value that is “true” if there is at least one location system that contains this component; for more details about location systems see Engine Locations from chapter 11.
- **componentImageId**: this is a number that represents an unique identifier of the component image; it is used in a special link to get the real image (http://www.haynespro-services.com/workshopServices3/getimage?id=)
- **genericParts**: this is a list of complex object containing details about the TecDoc Generic Articles that the component is equivalent to; the structure of these complex objects will be described next
- **linkedStories:** this is an array of LinkedStoriesV2 objects that contains additional information about the component
- **componentDescription:** this string contains the description of the component (for example: “Grounding point”)
- **componentCode:** the component code (for example “O8”)
- **symbolInstance:** the symbol instance of the component (for example “MC10”)

```
LinkedStoriesV2 {
      Integer            categoryId;
      String             categoryName;
      ExtStoryLineV5[]   stories;
}
```

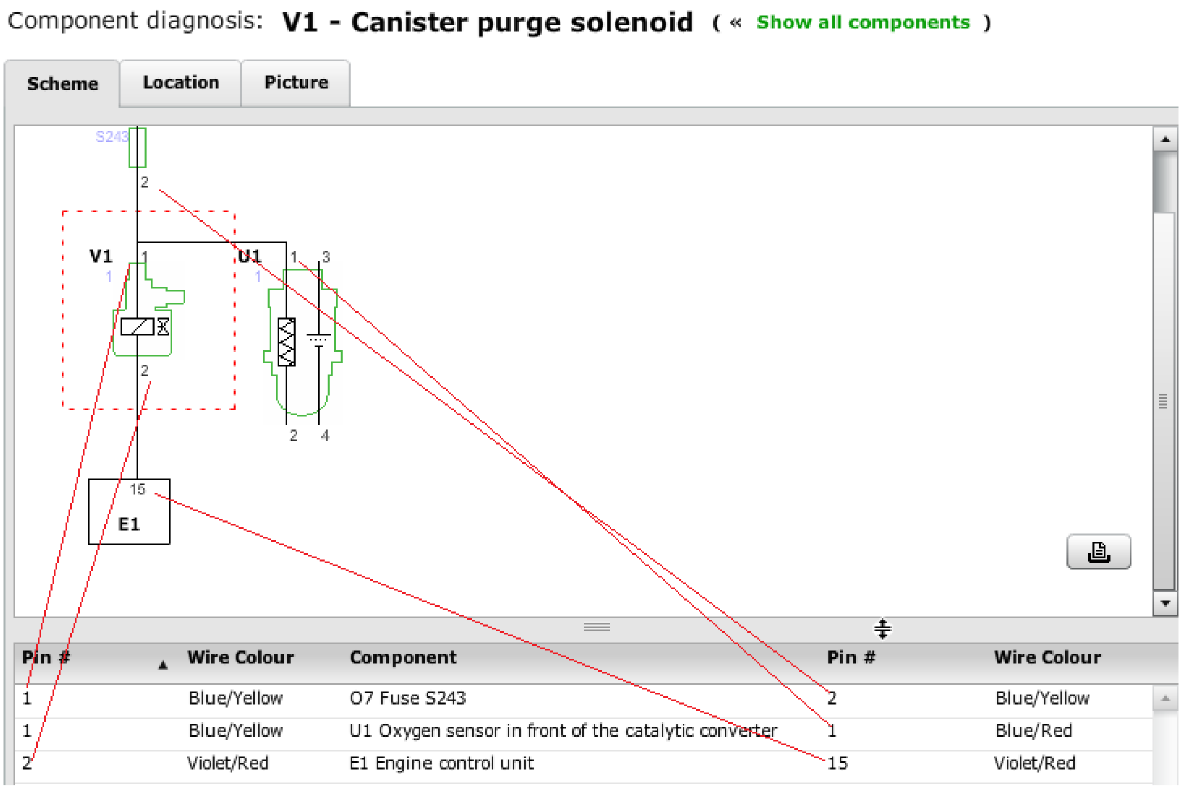
- **categoryId**: this property represents a unique identifier for the category of the repair manuals that are linked to a component.
- **categoryName**: represents name of the category.
- **stories:** this property is an array of ExtStoryLineV5 objects. The first level contains groups, and each group contain the second level in this property (subStoryLine) which contains the linked information about the component.

```
PinsArrayV2 {
      ComponentWsV2[]    componentsArray;
}
```

```
ComponentWsV2 {
      String       name;
      String       description;
      String       pinName;
      String       wireColor1;
      String       wireColor2;
```

```
        Boolean     hyperlink;
        int         si;
        int         st;
        String      pinNumber;
        Boolean     supplied;
        String      voltage;
}
```

- **name**: the code of the linked component (for example: “E1”)
- **description**: a text with the description of the linked component (for example: “Engine control unit”)
- **pinName**: the pin number of the requested component
- **wireColor1**: a text with the colour(s) of the wire that goes from the selected component to the the linked component
- **wireColor2**: a text with the colour(s) of the wire that goes from the linked component to the the current component (may be different from wireColor1 if there is also a connection between the two wires in the network)
- **hyperlink**: a boolean value which is “true” if the linked component can be shown in a scheme
- **si**: the symbol number id
- **st**: the system type id
- **pinNumber**: the pin number of the linked component
- **supplied :** This can take 3 values null (we show no value-arrow pair, false – we show voltage range value fallowed by a right arrow, true – we show left arrow fallowed by voltage range
- **voltage:** a range with minim and maxim voltage: ex 0.0 – 0.2V




```
WizardV3 {
        DiagnosisStepV3[]          diagnosisSteps;
        String                     title;
}
```

- **title**: this is a text containing the description of the diagnosis (for example: "Check the supply voltage (pin 1).")
- **diagnosisSteps**: this is a complex object that contain the data of the diagnosis; the structure of this object is described next

```
DiagnosisStepV3 {
        String                         type;
        String                         title;
        String                         text;
        String                         flashVarsString;
        ExtDiagramToolParameters       toolParameters;
        DiagnosisStepV3                yesAnswer;
        DiagnosisStepV3                noAnswer;
        String                         name;
        String                         wirediagramLink;
        Graph                          graph;
        Table                          table;
        String[]                       scopeImagesPaths;
        Integer                        sortOrder;
}
```

- **title**: this is the title of the diagnosis (for example: “Check the supply voltage (pin 1).”). This structure being a tree-structure, this will have a value only on the top-most level
- **text**: This is a text that describes the diagnosis step, what and how should be measured (for example: “*Check the following: [component si=106231930,st=2971]R3 Fuel pump relay[/component]*”, or “*Turn the ignition on, crank or start the engine. Measure the voltage on pin 1. Is it between 12 and 14.4 V?*”). The text can contain an xml-like tag using “[“ and “]” as tag-delimiters. The tag is intended as a link to a component (by calling the same operation with the si and st parameters specified in the tag)
- **type**: this is the type of diagnosis step; it can be one of the following:
  - "**yes/no question**" (eg. for: “*Turn the ignition on, crank or start the engine. Measure the voltage on pin 87 ([component si=106231930,st=2971,shield=106843114]R3 Fuel pump relay[/component]). Is it between 12 and 14.4 V?*”)
  - "**ok answer**" (eg. for: “*The fuse passed this test.*”)
  - "**fault answer**" (eg. for: “*Check the following: [component si=106231930,st=2971]R3 Fuel pump relay[/component].*”)
- **flashVarsString**: this contains information useful for representing the diagnostic step on the diagram (for example, to show the measurement equipment, the pins that the measurement is performed on or the value the measurement equipment should display;e.g.: “multimeter,normal,13.2 V,106231878 106842950”) - **Deprecated** (you can use the “toolParameters” parameter)
- **toolParameters**: this is a complex object that contains information useful for representing the diagnostic step on the digram (the same as flashVarsString, but structured). The structure of the toolParameters will be presented next.
- **yesAnswer**: it is a complex object with the same structure, which represents the next step for a “yes/no question” step in case the answer to the question is “yes”
- **noAnswer**: it is a complex object with the same structure, which represents the next step for a “yes/no question” step in case the answer to the question is “no”
- **name – not used**
- **sortOrder:** this is a number representing the sort order
- **table:** this is a complex object representing data in a table format (columns, rows, headers for each column). The exact structure will be presented next. The information in this object is in direct relation with the diagnostic step (for example, it can be a set of values based on which to determine if a component works correctly)
- **wirediagramLink**: this is a text representing a URL to a SVG file which represent the local diagram for the diagnostic step (it can be null if no diagram is required for that step)
- **graph**: this is a complex object representing a 2 dimensional graph (x, y axis). The data represented is in direct relation to the diagnostic step. The exact structure of this object will be described next
- **scopeImagesPaths**: this is a text representing a URL to a SVG file which represent the image of a wave form (oscilloscope); can be null;

```
Table {
      Row    header;
      Row[] rows;
}
```

- **header**: this is a complex object that contains the header for each column
- **rows**: this is a list of complex objects that contain the values corresponding to each row

```
Row {
      String value1;
      String value2;
}
```

- **value1**: this is an object containing a text value corresponding to column one
- **value2**: this is an object containing a text value corresponding to column two

```
Graph {
      String      xaxis;
      String      yaxis;
      Point[]     points;
}
```

- **xaxis**: this is an object containing the caption for the X-axis of the graph
- **yaxis**: this is an object containing the caption for the Y-axis of the graph
- **points**: this is a list of complex objects that contain the coordinates for each point or vertical bar of the graph

```
Point {
      double      xvalue;
      double      yminvalue;
      double      ymaxvalue;
}
```

- **xvalue**: the x coordinate
- **yminvalue**: the minimal value of the y coordinate
- **ymaxvalue**: the maximal value of the y coordinate

2.3.2 Operation: "getCompleteComponentV10"

This operation requires as parameters the car id, a component id, a system-type id, language and includeSmartLinks.

These are the request parameters and their meaning:

- typeId – int (this number is the "id" of the Type found in ExtCarType; you can check chapter 3 for more information about this object)
- language - String (this should be a 2 character string, established by the 639-1 ISO; for example, for English it is "en", for French it is "fr")
- si – a number, the id of the "component_id" of the SignalInstance object (see the 4.2 subchapter for more details)

- st – a number, the id of the system-type
  The difference between V8 and V7 is that the latest version localWiringImageExtended contains the parameter policy=full which is necessary for obtaining the extended full diagram Also a new parameter has been introduce in the response, pinTableExtended which contain the components from the newly extended diagram.
- includeSmartLinks – boolean value to show or not smart links linked to vesa components

  The **difference** between V10 and V9 is that we have introduced a new parameter *includeSmartLinks* and in response we added smart links for repair manuals linked to vesa components.The structure permits adding smart links from vesa components to other subjects in the future.

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:data="http://data.webservice.workshop.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <data:getCompleteComponentV10>
         <data:vrid>${vrid}</data:vrid>
         <data:typeId>318000062</data:typeId>
         <data:si>327887674</data:si>
         <data:st>319815489</data:st>
         <data:language>en</data:language>
         <data:includeSmartLinks>true</data:includeSmartLinks>
         </data:getCompleteComponentV10>
     </soapenv:Body>
  </soapenv:Envelope>
```

- response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getCompleteComponentV10Response
xmlns="http://data.webservice.workshop.vivid.nl">
         <getCompleteComponentV10Return
xmlns:ns1="http://datamodel.webservice.elecdata.vivid.nl">
            <ns1:completeComponentV10
xmlns:ns2="http://datatypes.webservice.elecdata.vivid.nl">
               <ns2:componentCode>U1</ns2:componentCode>
               <ns2:componentDescription>Oxygen sensor in front of the catalytic
converter</ns2:componentDescription>
               <ns2:componentImageId xsi:nil="true"/>
               <ns2:connectorSideImage xsi:nil="true"/>
  ..................
  <ns2:pinTableExtended>
  ............
   </ns2:pinTableExtended>
```

```
                    <ns2:renderable>true</ns2:renderable>
                    <ns2:smartLinks>
                       <item>
                          <filter xsi:nil="true"/>
                          <id1>319016956</id1>
                          <id2 xsi:nil="true"/>
                          <operation>getStoryInfo</operation>
                          <text>
                             <item>Injector: removal/installation/setting</item>
                          </text>
                       </item>
                    </ns2:smartLinks>
                    <ns2:symbolInstance>1</ns2:symbolInstance>
                 </ns1:completeComponentV10>
```

The **difference** between V9 and V8 is that for the pin table and pin table extended we have introduced 2 new parameters: supplied (This can take 3 values *null* (we show no value-arrow pair, *false* – we show voltage range value fallowed by a right arrow, *true* – we show left arrow fallowed by voltage range value) and voltage: this is String value representing the voltage in or out the component (usually is an ECU component)

Example of E1 (ECU) component:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:data="http://data.webservice.workshop.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <data:getCompleteComponentV10>
         <data:vrid>${vrid}</data:vrid>
         <data:typeId>102002271</data:typeId>
         <data:si>334234567</data:si>
         <data:st>320387431</data:st>
         <data:language>en</data:language>
          </data:getCompleteComponentV10>
      </soapenv:Body>
   </soapenv:Envelope>
```

- response:

```
   <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
      <soapenv:Body>
         <getCompleteComponentV10Response
xmlns="http://data.webservice.workshop.vivid.nl">
            <getCompleteComponentV10Return
xmlns:ns1="http://datamodel.webservice.elecdata.vivid.nl">
```

```
            <ns1:completeComponentV10 xmlns:ns2="http://datatypes.webservice.elecdata.vivid.nl">
                <ns2:componentCode>E1</ns2:componentCode>
                <ns2:componentDescription>Engine control unit</ns2:componentDescription>
                <ns2:componentImageId>1070</ns2:componentImageId>
                <ns2:connectorSideImage>https://acc.haynespro-assets.com/workshop/images/elecdata/301000169.svgz</ns2:connectorSideImage>
                <ns2:diagnosisSteps>
 …
    <ns2:pinTable>
                    <ns2:componentsArray>
                        <item>
                            <ns2:description>Engine control unit</ns2:description>
                            <ns2:hyperlink>true</ns2:hyperlink>
                            <ns2:name>E1</ns2:name>
                            <ns2:pinName>A 5</ns2:pinName>
                            <ns2:pinNumber>A 1</ns2:pinNumber>
                            <ns2:si>334234567</ns2:si>
                            <ns2:st>320387431</ns2:st>
                            <ns2:supplied xsi:nil="true"/>
                            <ns2:voltage xsi:nil="true"/>
                            <ns2:wireColor1>Pink</ns2:wireColor1>
                            <ns2:wireColor2>Green</ns2:wireColor2>
                        </item>
                         …
                         <item>
                            <ns2:description>Grounding point GUD01</ns2:description>
                            <ns2:hyperlink>true</ns2:hyperlink>
                            <ns2:name>O8</ns2:name>
                            <ns2:pinName>1</ns2:pinName>
                            <ns2:pinNumber>A 4</ns2:pinNumber>
                            <ns2:si>334234641</ns2:si>
                            <ns2:st>320387431</ns2:st>
                            <ns2:supplied>true</ns2:supplied>
                            <ns2:voltage>0.0 - 0.2V</ns2:voltage>
                            <ns2:wireColor1>Black</ns2:wireColor1>
                            <ns2:wireColor2>Black</ns2:wireColor2>
                        </item>
                            …
                            <item>
                            <ns2:description>Main relay</ns2:description>
                            <ns2:hyperlink>true</ns2:hyperlink>
                            <ns2:name>R1</ns2:name>
                            <ns2:pinName>85</ns2:pinName>
                            <ns2:pinNumber>A 47</ns2:pinNumber>
                            <ns2:si>334234577</ns2:si>
                            <ns2:st>320387431</ns2:st>
                            <ns2:supplied>false</ns2:supplied>
                            <ns2:voltage>0.0 - 0.2V</ns2:voltage>
                            <ns2:wireColor1>White</ns2:wireColor1>
```

```
                              <ns2:wireColor2/>
                           </item>
…
```

How we show supplied and voltage in Pin Table:

a) Pin Number A1, Pin Name A5 – supplied = null, voltage = null

Pin number	Voltage	Wire colour/code	Component	Pin number	Wire colour/code
A 1		Pink	E1 Engine control unit	A 5	Green

b) Pin Number A4, Pin Name 1 – supplied = true, voltage = 0.0 – 0.02V

A 4	← 0.0 - 0.2V	Black	O8 Grounding point GUD01	1	Black

c) Pin Number A47, Pin Name 85 – supplied = false, voltage = 0.0 – 0.2V

A 47	0.0 - 0.2V →	White	R1 Main relay	85	

Pin number	Voltage	Wire colour/code	Component	Pin number	Wire colour/code
A 1		Pink	E1 Engine control unit	A 5	Green
A 1		Pink	E1 Engine control unit	A 3	Orange
A 1		Pink	O7 Fuse F38 EMS	2	
A 2		Black	E1 Engine control unit	A 4	Black
A 2	← 0.0 - 0.2V	Black	O8 Grounding point GUD01	1	Black
A 3		Orange	E1 Engine control unit	A 1	Pink
A 3		Orange	E1 Engine control unit	A 5	Green
A 3		Orange	O7 Fuse F38 EMS	2	
A 4		Black	E1 Engine control unit	A 2	Black
A 4	← 0.0 - 0.2V	Black	O8 Grounding point GUD01	1	Black
A 5		Green	E1 Engine control unit	A 1	Pink
A 5		Green	E1 Engine control unit	A 3	Orange
A 5		Green	O7 Fuse F38 EMS	2	
A 6	← 0.0 - 0.2V	Black	O8 Grounding point GUD02	1	Black

Partial pin table for this example from *TOUCH* (See **Voltage** column)

Example of non ECU component::

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <data:getCompleteComponentV10>
         <data:typeId>26650</data:typeId>
         <data:si>301248717</data:si>
         <data:st>2972</data:st>
         <data:language>en</data:language>
      </data:getCompleteComponentV10>
   </soapenv:Body>
</soapenv:Envelope>
```

- response (fragments):
  - fragment1:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getCompleteComponentV10Response xmlns="http://data.webservice.workshop.vivid.nl">
         <getCompleteComponentV10Return>
            <completeComponentV10>
               <componentImageId>1105</componentImageId>
               <connectorSideImage xsi:nil="true"/>
               <diagnosisSteps>
                  ...
                  </diagnosisSteps>
                  <ecuSideImage xsi:nil="true"/>
                  <genericParts xsi:nil="true"/>
                  <hasLocationSystems>false</hasLocationSystems>
                  <localWiringImageConcise><![CDATA[https://haynespro-services.com/workshopServices3/getSubnetDiagram?typeId=26650&si=301248717&st=2972&language=en&formatName=svg&vrid=HAPROXY-JSON-TEST&policy=localSubjectPolicy]]></localWiringImageConcise>
                  <localWiringImageExtended><![CDATA[https://haynespro-services.com/workshopServices3/getSubnetDiagram?typeId=26650&si=301248717&st=2972&language=en&formatName=svg&vrid=HAPROXY-JSON-TEST&policy=full]]></localWiringImageExtended>
                  <pinTable>
                     ...
                  </pinTable>
                  <pinTableExtended>
                    ...
                  </pinTableExtended>
                  <renderable>true</renderable>
                  ...
               </completeComponentV10>
               <status xsi:nil="true"/>
            </getCompleteComponentV10Return>
         </getCompleteComponentV10Response>
      </soapenv:Body>
   </soapenv:Envelope>
```

- fragment2:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getCompleteComponentV10Response xmlns="http://webservice.elecdata.vivid.nl">
         <getCompleteComponentV10Return>
            <completeComponentV10>
               <componentImageId>1105</componentImageId>
               <connectorSideImage xsi:nil="true"/>
               <diagnosisSteps>...</diagnosisSteps>
```

```
                  <ecuSideImage xsi:nil="true"/>
                      <genericParts xsi:nil="true"/>
                      <hasLocationSystems>false</hasLocationSystems>
                      <pinTable>
                         <componentsArray>
                            <item>
                               <description>Engine control unit</description>
                               <hyperlink>true</hyperlink>
                               <name>E1</name>
                               <pinName>64</pinName>
                               <pinNumber>1</pinNumber>
                               <si>106231877</si>
                               <st>2972</st>
                               <wireColor1>Violet/Red</wireColor1>
                               <wireColor2>Violet/Red</wireColor2>
                            </item>
                            ...
                         </componentsArray>
                      </pinTable>
                      <pinTableExtended>
                         <componentsArray>
                            <item>
                               <description>Fuse S10</description>
                               <hyperlink>true</hyperlink>
                               <name>O7</name>
                               <pinName/>
                               <pinNumber>1</pinNumber>
                               <si>301248761</si>
                               <st>2972</st>
                               <ns2:supplied xsi:nil="true"/>
                               <ns2:voltage xsi:nil="true"/>
                               <wireColor1>Red/Violet</wireColor1>
                               <wireColor2>Red/Green</wireColor2>
                            </item>
                           …
                       </pinTableExtended>
                      <renderable>true</renderable>
                      <subnetXml>...</subnetXml>
                   </completeComponentV10>
                   <status xsi:nil="true"/>
                </getCompleteComponentV10Return>
             </getCompleteComponentVResponse>
          </soapenv:Body>
       </soapenv:Envelope>
```

Pin #	Wire Colour	Component	Pin #	Wire Colour
1	Violet/Red	E1 Engine control unit	64	Violet/Red
1	Violet/Red	H3 Throttle control motor with position sensor	4	Violet/Red
2	Violet/Yellow	E1 Engine control unit	76	Violet/Yellow
3	Brown/Blue	E1 Engine control unit	67	Brown/Blue
3	Brown/Blue	H3 Throttle control motor with position sensor	7	Brown/Blue
3	Brown/Blue	L4 MAP and air temperature sensor	1	Brown/Blue
3	Brown/Blue	T1 Coolant temperature sensor	4	Brown/Blue
3	Brown/Blue	X1 Magnetic pick-up on the crankshaft	3	Brown/Blue

- fragment3:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getCompleteComponentV10Response xmlns="http://webservice.elecdata.vivid.nl">
         <getCompleteComponentV10Return>
            <completeComponentV10>
               <componentImageId>1105</componentImageId>
               <connectorSideImage xsi:nil="true"/>
               <diagnosisSteps>
                  <item>
                         <diagnosisSteps>
                            <item>
                               ...
                               <text>Turn the ignition on.         Measure the voltage on pin 3. Is it between 0 and 0.2 V?</text>
                               <title>Check the connection to ground (pin 3).</title>
                               <type>yes/no question</type>
                               <wirediagram2 xsi:nil="true"/>
                               <noAnswer>
                            ...
                                  <text>Turn the ignition on.         Measure the voltage on pin 67 ([component si=106231877,st=2972,shield=106843040]E1 Engine control unit[/component]). Is it between 0 and 0.2 V?</text>
                                  ...
                                  <yesAnswer>
                                     ...
                                     <text>Measure the resistance between pins 3 and 67 ([component si=106231877,st=2972,shield=106843040]E1 Engine control unit[/component]). The resistance should be less than 1 ohm. Check all wires and replace if necessary. See the diagram below for details on wire colours, connectors, welds and locations (if applicable).</text>
                                     ...
                                     <wirediagramLink>http://www.haynespro-services.com/workshopServices3/getLocalWiringDiagram?pid=106840283&cpid=106840221&connection=106461828&language=en&vrid=1671D4894D2C5B3B29B4D402FFE2574E&formatN
```

```
ame=svg</wiringdiagramLink>
                                    <yesAnswer xsi:nil="true"/>
                                 </yesAnswer>
                              </noAnswer>

                              ...
                           </item>
                           ...
                  </diagnosisSteps>
                  ...
               </completeComponentV10>
               <status xsi:nil="true"/>
            </getCompleteComponentV10Return>
         </getCompleteComponentV10Response>
      </soapenv:Body>
   </soapenv:Envelope>
```

HaynesPro®

X5 - Hall/MRE sensor on the camshaft

1: Check the connection to ground (pin 3).

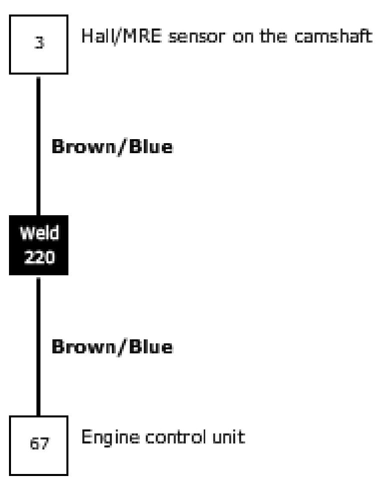
Turn the ignition on. Measure the voltage on pin 3. Is it between 0 and 0.2 V?

- ( ) Yes
- (•) No

Turn the ignition on. Measure the voltage on pin 67 (E1 Engine control unit). Is it between 0 and 0.2 V?

- (•) Yes
- ( ) No

Measure the resistance between pins 3 and 67 (E1 Engine control unit). The resistance should be less than 1 ohm. Check all wires and replace if necessary. See the diagram below for details on wire colours, connectors, welds and locations (if applicable).



2: Check the connectivity of pin 2.

- fragment4:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getCompleteComponentV10Response xmlns="http://webservice.elecdata.vivid.nl">
         <getCompleteComponentV10Return>
            <completeComponentV10>
               <componentImageId>1105</componentImageId>
               <connectorSideImage xsi:nil="true"/>
               <diagnosisSteps>...</diagnosisSteps>
               <ecuSideImage xsi:nil="true"/>

                        …
                  <linkedStories>
                        <linkedStories>
                           <categoryId>305000005</categoryId>
                           <categoryName>Component information</categoryName>
                           <stories>
                              <stories>
                                 <id xsi:nil="true"/>
                                 <mimeData xsi:nil="true"/>
                                 <name>Identification</name>
                                 <order>0</order>
                                 <paragraphContent xsi:nil="true"/>
                                 <remark xsi:nil="true"/>
                                 <sentenceGroupType xsi:nil="true"/>
                                 <sentenceStyle xsi:nil="true"/>
                                 <smartLinks xsi:nil="true"/>
                                 <specialTool xsi:nil="true"/>
                                 <status xsi:nil="true"/>
                                 <subStoryLines>
                                    <subStoryLines>
                                       <id xsi:nil="true"/>
                                       <mimeData xsi:nil="true"/>
                                       <name xsi:nil="true"/>
                                       <order>0</order>
                                       <paragraphContent>
                                          <paragraphContent>
                                             <id>306000817</id>
                                             <name>The operating principle is the same, irrespective of component design</name>
                                             <order>0</order>
<sentenceStyle>SENTENCE_STYLE</sentenceStyle>
                                          </paragraphContent>
                                          <paragraphContent>
                                             <id>306001362</id>
                                             <name>Camshaft Hall Effect pick-up sensors</name>
                                             <order>1</order>

```

```
                <sentenceStyle>SENTENCE_STYLE</sentenceStyle>
              </paragraphContent>
            </paragraphContent>
            <remark xsi:nil="true"/>
            <sentenceGroupType>PARAGRAPH</sentenceGroupType>
            <sentenceStyle xsi:nil="true"/>
            <smartLinks xsi:nil="true"/>
            <specialTool xsi:nil="true"/>
            <status xsi:nil="true"/>
            <subStoryLines xsi:nil="true"/>
            <tableContent xsi:nil="true"/>
          </subStoryLines>
        </subStoryLines>
        <tableContent xsi:nil="true"/>
      </stories>
      <stories>
        <id xsi:nil="true"/>
        <mimeData xsi:nil="true"/>
        <name>Function</name>
        <order>1</order>
        <paragraphContent xsi:nil="true"/>
        <remark xsi:nil="true"/>
        <sentenceGroupType xsi:nil="true"/>
        <sentenceStyle xsi:nil="true"/>
        <smartLinks xsi:nil="true"/>
        <specialTool xsi:nil="true"/>
        <status xsi:nil="true"/>
        <subStoryLines>
          <subStoryLines>
            <id xsi:nil="true"/>
            <mimeData xsi:nil="true"/>
            <name xsi:nil="true"/>
            <order>0</order>
            <paragraphContent>
              <paragraphContent>
                <id>306000847</id>
                <name>The sensor uses a special semiconductor to convert a changing magnetic field into a changing voltage</name>
                <order>0</order>
                <sentenceStyle>SENTENCE_STYLE</sentenceStyle>
              </paragraphContent>
              <paragraphContent>
                <id>306000848</id>
                <name>A stronger magnetic field produces a greater voltage</name>
                <order>1</order>
                <sentenceStyle>SENTENCE_STYLE</sentenceStyle>
              </paragraphContent>
    ...
```

2.4 Component List

2.4.1 Structure

ListComponentArrayContainer {
    ListComponentArray[]   listComponentArray;
}

ListComponentArray {
    int   system_type_id;
    String   system_text;
    ListComponent[]   components;
}

- **system_type_id**: a number, a unique identifier for the system type
- **system_text**: this is a text that describes the system
- **components**: a list of complex objects representing the components of the system

ListComponent {
    int   si;
    int   st;
    String   text;
}

- **si**: this is a number that represents the id of the component
- **st**: a number, a unique identifier for the system type (the same as the “system_type_id” from the ListComponentArray)
- **text**: this is a text containing the description of the component

2.4.2 Operation: getComponentLists

This operation returns the full list of components in the selected system or systems, and it requires as parameters the system type id(s) and the language.

These are the request parameters and their meaning:

- st – a number, the id of the system-types
- language - String (this should be a 2 character string, established by the 639-1 ISO; for example, for English it is “en”, for French it is “fr”)

Example:

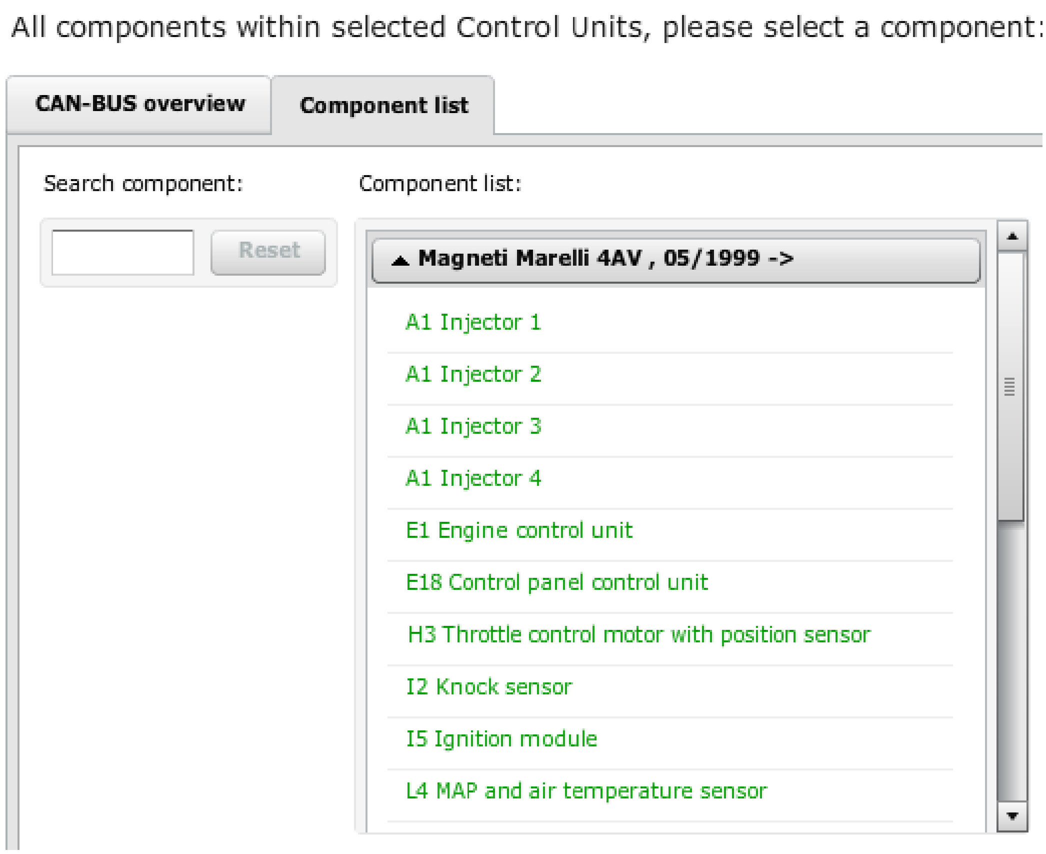
- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:getComponentLists>
         <!--1 or more repetitions:-->
```

```
        <web:system_types_ids>2972</web:system_types_ids>
        <web:language>en</web:language>
     </web:getComponentLists>
  </soapenv:Body>
</soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <soapenv:Body>
     <getComponentListsResponse xmlns="http://webservice.elecdata.vivid.nl">
        <getComponentListsReturn>
           <listComponentArray>
              <item>
                 <components>
                    <item>
                       <si>106231920</si>
                          <st>2972</st>
                          <text>A1 Injector 1</text>
                       </item>
                       <item>
                          <si>106231913</si>
                          <st>2972</st>
                          <text>A1 Injector 2</text>
                       </item>
                       ...
                    </components>
                    <system_text>Magneti Marelli 4AV, 05/1999 -></system_text>
                    <system_type_id>2972</system_type_id>
                 </item>
              </listComponentArray>
              <status xsi:nil="true"/>
           </getComponentListsReturn>
        </getComponentListsResponse>
     </soapenv:Body>
  </soapenv:Envelope>
```



2.4.3 Calling getGeneralArticlesByComponentNumber()

This operation requires a set of component codes and returns the genarts (TecDoc general article numbers) for each of them.

The structure of the returned objects is:

```
ExtComponentGenrats  {
        String                              componentNumber;
        ExtGeneralArticleDescription[]      generalArticles;
}
```

- **componentNumber**: this is a text that represent the component number (from the parameter list) for which the general articles are listed

```
ExtGeneralArticleDescription {
        Integer     id;
        String      description;
}
```

- **id**: the TecDoc genart
- **description**: this is a text with the description (language dependent) of the general article number from TecDoc (for example: "injector");

The required parameters are:

- descriptionLanguage - String (this should be a 2 character string, established by the 639-1 ISO; for example, for English it is “en”, for French it is “fr”)
- componentNumber – String[] (this is a list of text elements that represent component codes)

This a sample request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:data="http://data.webservice.workshop.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <data:getGeneralArticlesByComponentNumber>
         <data:vrid>[please enter your vrid here]</data:vrid>
         <data:descriptionLanguage>en</data:descriptionLanguage>
         <!--1 or more repetitions:-->
         <data:componentNumber>E1</data:componentNumber>
         <data:componentNumber>H1</data:componentNumber>
            <data:componentNumber>X1</data:componentNumber>
         </data:getGeneralArticlesByComponentNumber>
      </soapenv:Body>
   </soapenv:Envelope>
```

And this is a fragment of the response for the previous request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getGeneralArticlesByComponentNumberResponse
xmlns="http://data.webservice.workshop.vivid.nl">
         <getGeneralArticlesByComponentNumberReturn>
            <componentNumber>E1</componentNumber>
            <generalArticles>
               <item>
                  <description>Control unit, engine management</description>
                  <id>3956</id>
                     <status xsi:nil="true"/>
                  </item>
               </generalArticles>
               <status xsi:nil="true"/>
            </getGeneralArticlesByComponentNumberReturn>
            ...
            <getGeneralArticlesByComponentNumberReturn>
               <componentNumber>X1</componentNumber>
               <generalArticles>
                  <item>
                     <description>Sensor, crankshaft pulse</description>
                     <id>833</id>
```

```
                  <status xsi:nil="true"/>
               </item>
               <item>
                  <description>Sensor, ignition pulse</description>
                  <id>834</id>
                  <status xsi:nil="true"/>
               </item>
               <item>
                  <description>Rpm sensor, engine management</description>
                  <id>3896</id>
                  <status xsi:nil="true"/>
               </item>
            </generalArticles>
            <status xsi:nil="true"/>
         </getGeneralArticlesByComponentNumberReturn>
      </getGeneralArticlesByComponentNumberResponse>
   </soapenv:Body>
</soapenv:Envelope>
```

2.5 Fault Codes

2.5.1 Structure

```
FaultCode3ArrayContainer {
        FaultCode3[]  faultCode3Array;
}

FaultCode3  {
        Integer                  id;
        String                   fc_code;
        String                   fc_code2;
        Integer                  fd_component_name_id;
        Integer                  fd_si_number;
        Boolean                  fd_plausible;
        Integer                  fd_id;
        String                   ft_text;
        String                   ct_text;
        DiagnosticComponentV2[]  diagnosticComponents;
}
```

- **id**: this is a number that uniquely identifies the FaultCode2 object
- **fc_code**: this is a text containing the code (e.g.: “P3358”)
- **fc_code2**: this is a text containing a code; it is usually null; when it is present, it represents an alternative code equivalent to “fc_code” (for example, fc_code is “P0004” while fc_code is “16388” for “Fuel delivery control solenoid circuit high”)
- **fd_component_name_id**: not used
- **fd_si_number**: not used
- **fd_plausible**: not used
- **fd_id**: not used
- **ft_text**: a text with the description of the fault code
- **ct_text**: a text with the description of the involved component or components (if more than one, the text will contain each one on a different row)
- **diagnosticComponents**: this is a list of complex objects that are involved in the diagnostic process for the corresponding fault code; it can be null; the structure of these objects is presented next

```
FaultCode4ArrayContainer {
        FaultCode3[] faultCode3Array;
        TSBArrayContainer[] tsbArrayContainer;
}
```

TSBArrayContainer{
```
String faultCode;
ExtTSBCategoryV4[] extTsbCategoryArray;
```
}

- **faultCode**: this is the searched fault code
- **extTsbCategoryArray:** this is an array of tsb categories, each category (SmartCase, SmartFix, Recall) containing an array of linked systems.For more information about what this object means please consult the other documentation Data_Exchange_Web-Services_3

FaultCode5ArrayContainer {
FaultCode4[] faultCode4Array;
TSBArrayContainer[] tsbArrayContainer;
}

FaultCode4{
```
Integer                      id;
String[]                     faultCodes;
Integer                      fd_component_name_id;
Integer                      fd_si_number;
Boolean                      fd_plausible;
Integer                      fd_id;
String                       ft_text;
String                       ct_text;
DiagnosticComponentV2[]      diagnosticComponents;
```
}

This structure is similar to FaultCode3 with the difference that fc_cocde and fc_code2 are now sent in faultCodes array together with a fc_code3.This array contains the all possible synonyms for a searched fault code.

DiagnosticComponentV2 {
```
Integer        koppeltype;
ComponentV2    component;
Integer        system_type_id;
Integer        system_group_id;
Integer        sort_order;
```
}

- **kooppeltype**: is a number which can have values 1 or 2:
  - 1 for Component Koopeling
  - 2 for Component Diagnostic Step
- **component**: this is a complex object representing a component
- **system_type_id**: this is the identifier of the system type
- **system_group_id**: this is the identifier of system group
- **sort_order**: this is a number representing the sort order of the DiagnosticComponent objects

```
ComponentV2 {
        String                  name;
        String                  number;
        Integer                 symbolInstanceId;
        Integer                 systemTypeId;
        Integer                 shieldPinId;
        Integer                 slImagesId;
        Integer                 loX;
        Integer                 loY;
        String                  loDirection;
        ExtGeneralArticle[]     generalArticles;
}
```

- **name**: this is the name of the component
- **symbolInstanceId**: this is an identifier of the component
- **systemTypeId**: this is an identifier of the system type
- **shieldPinId**: not used
- **slImagesId**: not used
- **loX**: not used
- **loY**: not used
- **loDirection**: not used
- **number**: the component code (for example “A3”)
- **generalArticles**: this is a list of general article numbers (with TecDoc ids)

2.5.2 Operation: getFaultCodesForType

This operation returns all the known fault codes that can appear for a car type. It requires the car type identifier and the language as parameters and it returns an object with the structure described above.

This is an operation which returns a very large amount of data and it requires a larger amount of time to retrieve this data. For this reason, this operation is not used in the existing implementation of the graphical interface.

2.5.3 Operation: filterFaultCodesForType

This operation requires as parameter a car type identifier and a list of fault codes. It returns all the known details (in the format described above).

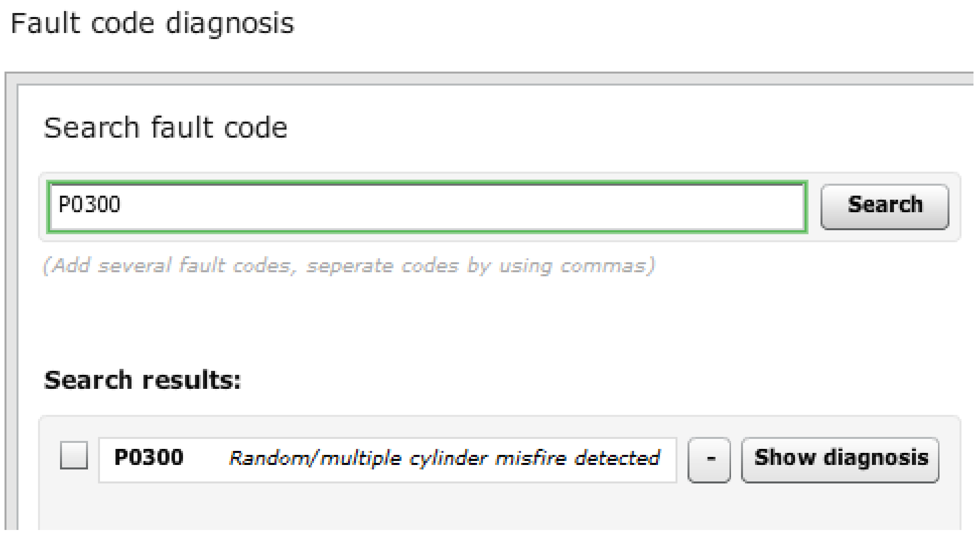
Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:filterFaultCodesForType>
         <web:typeId>26650</web:typeId>
         <!--1 or more repetitions:-->
         <web:codesin>P0330</web:codesin>
         <web:language>en</web:language>
      </web:filterFaultCodesForType>
   </soapenv:Body>
</soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <filterFaultCodesForTypeResponse
xmlns="http://webservice.elecdata.vivid.nl">
         <filterFaultCodesForTypeReturn>
            <faultCode2Array>
               <item>
                  <ct_text>Knock sensor
Ignition module
Ignition coil
Spark plug
Hall/MRE sensor on the crankshaft
Magnetic pick-up on the crankshaft</ct_text>
                     <diagnosticComponents xsi:nil="true"/>
                     <fc_code>P0330</fc_code>
                     <fc_code2 xsi:nil="true"/>
                     <fc_fault_code_system_id xsi:nil="true"/>
                     <fd_component_name_id xsi:nil="true"/>
                     <fd_id xsi:nil="true"/>
                     <fd_plausible xsi:nil="true"/>
                     <fd_si_number xsi:nil="true"/>
                     <ft_text>Knock sensor 2 circuit</ft_text>
                     <id>337</id>
                  </item>
               </faultCode2Array>
               <status xsi:nil="true"/>
            </filterFaultCodesForTypeReturn>
         </filterFaultCodesForTypeResponse>
      </soapenv:Body>
   </soapenv:Envelope>
```



2.5.4 Operation: filterFaultCodesWithDiagnosticStepsForTypeV2

This operation requires the same parameters as “filterFaultCodesForType”, but it returns as extra information the diagnostic steps. Also, if there are more fault-codes entered, the last element is a combination of all the codes with the diagnisis ordered based on the probability of the component of being the cause of the fault.

Example:

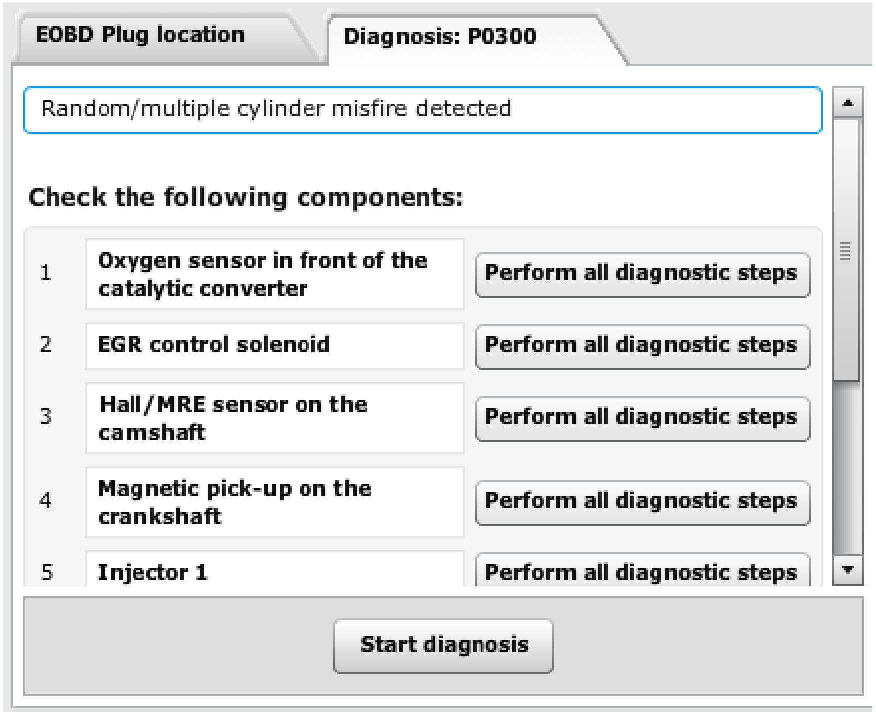
- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:filterFaultCodesWithDiagnosticStepsForType>
         <web:vrid>?</web:vrid>
         <web:typeId>26650</web:typeId>
         <!--1 or more repetitions:-->
         <web:codesin>P0300</web:codesin>
         <web:language>en</web:language>
          </web:filterFaultCodesWithDiagnosticStepsForType>
      </soapenv:Body>
  </soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <filterFaultCodesWithDiagnosticStepsForTypeV2Response
xmlns="http://webservice.elecdata.vivid.nl">
         <filterFaultCodesWithDiagnosticStepsForTypeV2Return>
            <faultCode2Array>
```

```
                <item>
                   <ct_text>EGR control solenoid
Injector
Oxygen sensor in front of the catalytic converter
Ignition coil
Hall/MRE sensor on the crankshaft
Hall/MRE sensor on the inlet camshaft
Hall/MRE sensor on the camshaft
EGR control solenoid with position sensor
Magnetic pick-up on the crankshaft
Spark plug</ct_text>
                      <diagnosticComponents>
                         <item>
                            <component>
                               <loDirection xsi:nil="true"/>
                               <loX xsi:nil="true"/>
                               <loY xsi:nil="true"/>
                               <name>Oxygen sensor in front of the catalytic converter</name>
                               <shieldPinId xsi:nil="true"/>
                               <slImagesId xsi:nil="true"/>
                               <symbolInstanceId>106231931</symbolInstanceId>
                               <systemTypeId>2972</systemTypeId>
                            </component>
                            <couplingType>1</couplingType>
                            <system_group_id>2</system_group_id>
                            <system_type_id>2972</system_type_id>
                         </item>
                         ...
                      </diagnosticComponents>
                      <fc_code>P0300</fc_code>
                      <fc_code2 xsi:nil="true"/>
                      <fc_fault_code_system_id xsi:nil="true"/>
                      <fd_component_name_id xsi:nil="true"/>
                      <fd_id xsi:nil="true"/>
                      <fd_plausible xsi:nil="true"/>
                      <fd_si_number xsi:nil="true"/>
                      <ft_text>Random/multiple cylinder misfire detected</ft_text>
                      <id>305</id>
                   </item>
                </faultCode2Array>
                <status xsi:nil="true"/>
             </filterFaultCodesWithDiagnosticStepsForTypeV2Return>
          </filterFaultCodesWithDiagnosticStepsForTypeV2Response>
       </soapenv:Body>
    </soapenv:Envelope>
```



2.5.5 Operation: filterFaultCodesWithDiagnosticStepsForTypeV3

This operation has the same parameters as filterFaultCodesWithDiagnosticStepsForTypeV2, but it returns as extra information the tsb systems linked to the searched fault codes. These systems are used in workshop online application to show smart links from fault code to tsb, recalls or smart cases.

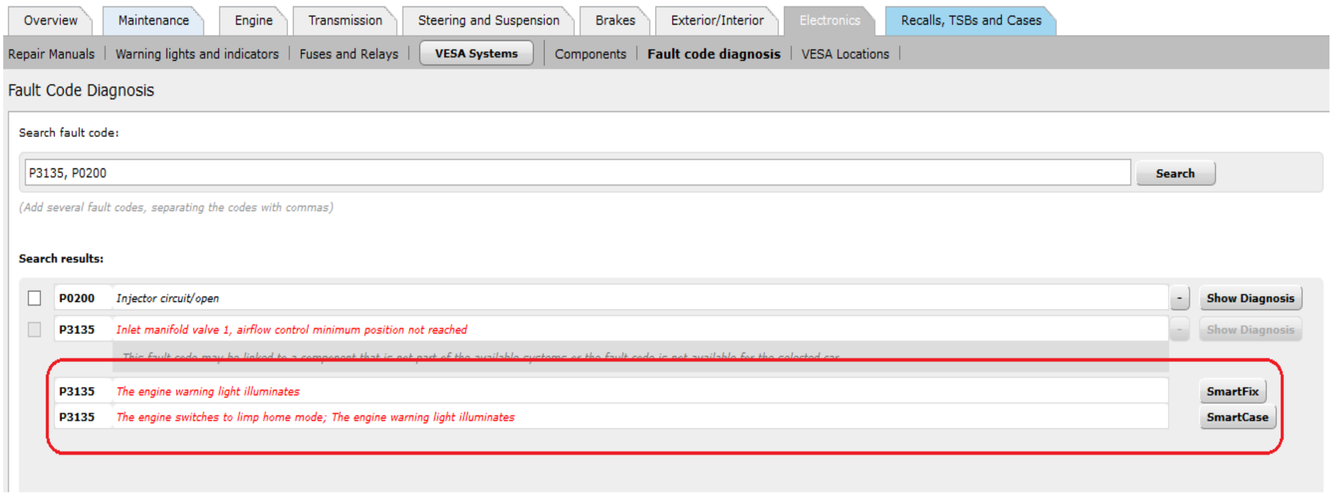


Figure 4.5.5 Workshop Online – TSB smart links

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:data="http://data.webservice.workshop.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <data:filterFaultCodesWithDiagnosticStepsForTypeV3>
         <data:vrid>?</data:vrid>
         <data:typeId>72910</data:typeId>
         <!--1 or more repetitions:-->
         <data:codesin>P3135</data:codesin>
         <data:language>en</data:language>
         </data:filterFaultCodesWithDiagnosticStepsForTypeV3>
      </soapenv:Body>
   </soapenv:Envelope>
```

response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <filterFaultCodesWithDiagnosticStepsForTypeV3Response
xmlns="http://data.webservice.workshop.vivid.nl">
         <filterFaultCodesWithDiagnosticStepsForTypeV3Return>
            <faultCode3Array>
               <faultCode3Array>
                  <ct_text/>
                  <diagnosticComponents/>
                  <fc_code>P3135</fc_code>
                     <fc_code2>19591</fc_code2>
                     <fd_component_name_id xsi:nil="true"/>
```

```xml
                  <fd_id xsi:nil="true"/>
                  <fd_plausible xsi:nil="true"/>
                  <fd_si_number xsi:nil="true"/>
                  <ft_text>Inlet manifold valve 1, airflow control minimum position not reached</ft_text>
                  <id>104000841</id>
               </faultCode3Array>
            </faultCode3Array>
            <status xsi:nil="true"/>
            <tsbArrayContainer>
               <tsbArrayContainer>
                  <extTsbCategoryArray>
                     <extTsbCategoryArray>
                        <categoryDescription>SmartFIX</categoryDescription>
                        <categoryTypeConstant>TSB</categoryTypeConstant>
                        <status xsi:nil="true"/>
                        <systems>
                           <systems>
                              <criteria xsi:nil="true"/>
                              <issueDate>02/01/2013</issueDate>
                              <oeCode>2016504/22</oeCode>
                              <systemDescription>The engine warning light illuminates</systemDescription>
                              <systemId>301001242</systemId>
                           </systems>
                        </systems>
                     </extTsbCategoryArray>
                     <extTsbCategoryArray>
                        <categoryDescription>SmartCASE</categoryDescription>
                        <categoryTypeConstant>CASE</categoryTypeConstant>
                        <status xsi:nil="true"/>
                        <systems>
                           <systems>
                              <criteria xsi:nil="true"/>
                              <issueDate xsi:nil="true"/>
                              <oeCode xsi:nil="true"/>
                              <systemDescription>The engine switches to limp home mode; The engine warning light illuminates</systemDescription>
                              <systemId>301000430</systemId>
                           </systems>
                        </systems>
                     </extTsbCategoryArray>
                  </extTsbCategoryArray>
                  <faultCode>P3135</faultCode>
               </tsbArrayContainer>
            </tsbArrayContainer>
         </filterFaultCodesWithDiagnosticStepsForTypeV3Return>
      </filterFaultCodesWithDiagnosticStepsForTypeV3Response>
   </soapenv:Body>
</soapenv:Envelope>
```

2.5.6 Operation: filterFaultCodesWithDiagnosticStepsForTypeV4

This operation has the same parameters as filterFaultCodesWithDiagnosticStepsForTypeV3, but it returns as extra information the 3rd fault code synonym. These synonyms are shown into online application for the searched results. Ex: when we search P0300 this can have another 2 synonyms (16681 and 412C) and they are displayed on the same line:

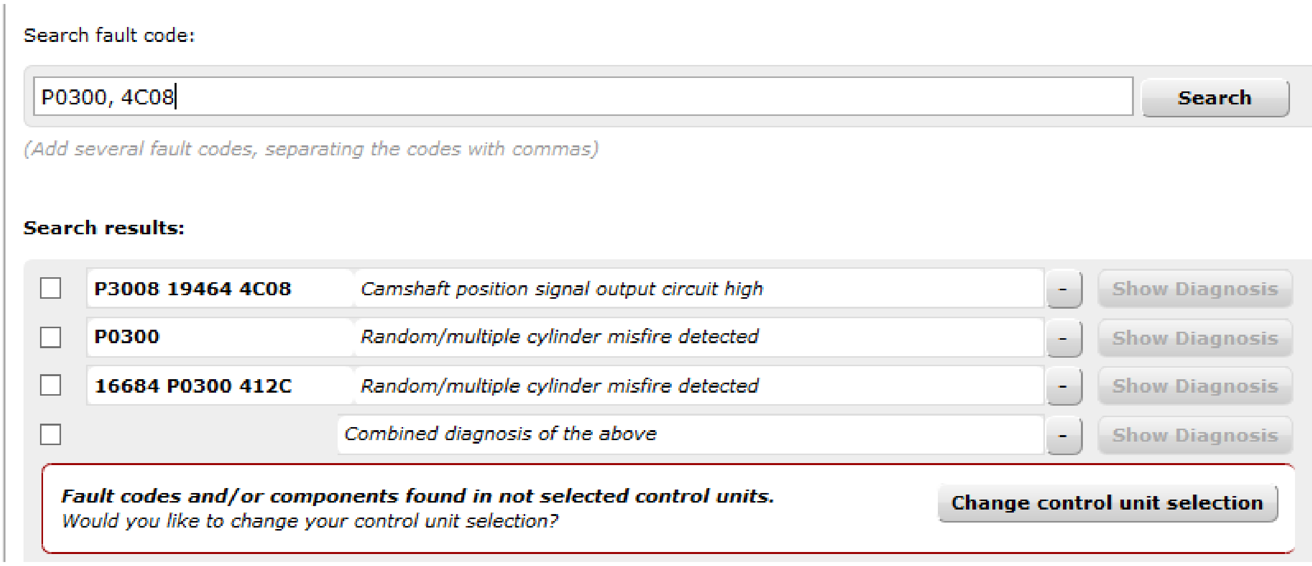


Figure 4.5.6 Workshop Online – fault codes search results

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:data="http://data.webservice.workshop.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <data:filterFaultCodesWithDiagnosticStepsForTypeV4>
         <data:vrid>?</data:vrid>
         <data:typeId>54200</data:typeId>
         <!--1 or more repetitions:-->
         <data:codesin>P0300</data:codesin>
         <data:language>en</data:language>
      </data:filterFaultCodesWithDiagnosticStepsForTypeV4>
   </soapenv:Body>
</soapenv:Envelope>
```

- response:

```
   <faultCodes>
                  <faultCodes>16684</faultCodes>
                  <faultCodes>P0300</faultCodes>
                  <faultCodes>412C</faultCodes>
```

```
                    </faultCodes>
                    <fd_component_name_id xsi:nil="true"/>
                    <fd_id xsi:nil="true"/>
                    <fd_plausible xsi:nil="true"/>
                    <fd_si_number xsi:nil="true"/>
                        <ft_text>Random/multiple cylinder misfire detected</ft_text>
                        <id>989</id>
```

The difference between the previous version and this is that instead of sending fc_code and fc_code2 parameters we send an array of fault codes synonyms for each searched fault code.
As we could see in the online example the search results provide an extra row for Combined diagnosis of the above. This result it is always the last result from calling this operation, but only when the length of the results is > 1 and we have searched at least 2 different fault codes.

2.5.7 Operation: getFaultCode2ForId

This operation requires the language and the identifier of a fault code object (as returned by getFaultCodesForType, filterFaultCodesForType or filterFaultCodesWithDiagnosticStepsForType).

Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:getFaultCode2ForId>
         <web:vrid>?</web:vrid>
         <web:faultcode>337</web:faultcode>
         <web:language>en</web:language>
      </web:getFaultCode2ForId>
   </soapenv:Body>
</soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getFaultCode2ForIdResponse xmlns="http://webservice.elecdata.vivid.nl">
         <getFaultCode2ForIdReturn>
            <faultCode2>
               <ct_text>Knock sensor
Ignition module
Ignition coil
Spark plug
Hall/MRE sensor on the crankshaft
Magnetic pick-up on the crankshaft</ct_text>
                  <diagnosticComponents xsi:nil="true"/>
```

```
                      <fc_code>P0330</fc_code>
                      <fc_code2 xsi:nil="true"/>
                      <fc_fault_code_system_id xsi:nil="true"/>
                      <fd_component_name_id xsi:nil="true"/>
                      <fd_id xsi:nil="true"/>
                      <fd_plausible xsi:nil="true"/>
                      <fd_si_number xsi:nil="true"/>
                      <ft_text>Knock sensor 2 circuit</ft_text>
                      <id>337</id>
                   </faultCode2>
                   <status xsi:nil="true"/>
                </getFaultCode2ForIdReturn>
             </getFaultCode2ForIdResponse>
          </soapenv:Body>
       </soapenv:Envelope>
```

2.5.8 Operation: getDiagnosticStepsForFaultCodeId

This operation requires the fault code object identifier and the system type identifier and it returns an diagnosticComponentArray object with the components involved in the diagnostic of the specified fault code.
Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:getDiagnosticStepsForFaultCodeId>
         <web:vrid>?</web:vrid>
         <web:faultcodeId>337</web:faultcodeId>
         <!--1 or more repetitions:-->
         <web:system_types_ids>2972</web:system_types_ids>
         <web:language>en</web:language>
          </web:getDiagnosticStepsForFaultCodeId>
       </soapenv:Body>
    </soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getDiagnosticStepsForFaultCodeIdResponse
xmlns="http://webservice.elecdata.vivid.nl">
         <getDiagnosticStepsForFaultCodeIdReturn>
            <diagnosticComponentArray>
               <item>
                  <component>
                     <loDirection xsi:nil="true"/>
                     <loX xsi:nil="true"/>
```

```
                     <loY xsi:nil="true"/>
                     <name>Magnetic pick-up on the crankshaft</name>
                     <shieldPinId xsi:nil="true"/>
                     <slImagesId xsi:nil="true"/>
                     <symbolInstanceId>106231929</symbolInstanceId>
                     <systemTypeId>2972</systemTypeId>
                  </component>
                  <couplingType>1</couplingType>
                  <system_group_id>2</system_group_id>
                  <system_type_id>2972</system_type_id>
               </item>
               ...
            </diagnosticComponentArray>
            <status xsi:nil="true"/>
         </getDiagnosticStepsForFaultCodeIdReturn>
      </getDiagnosticStepsForFaultCodeIdResponse>
   </soapenv:Body>
</soapenv:Envelope>
```

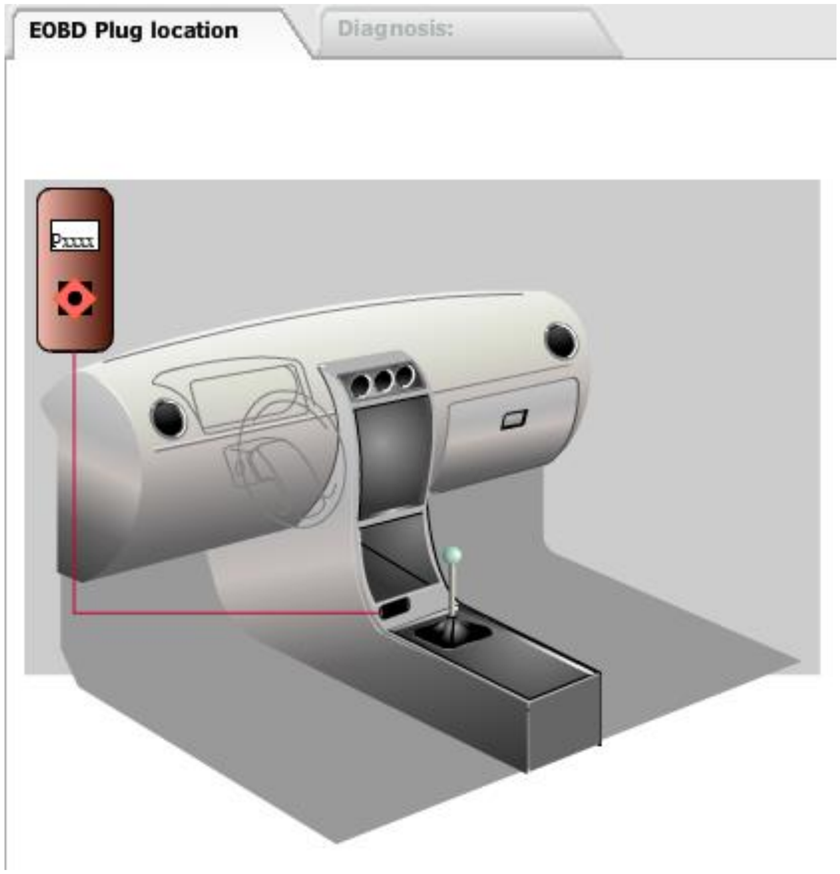
2.6 Diagnostic Connector

2.6.1 Structure

DiagnosisConnectorLocImageContainer {
    LocationImage   image;
}

LocationImage {
    String   imagePath;
    Integer   height;
    Integer   width;
    long   size;
}

- imagePath: this is a text containing the URL to the image
- height: this is a number representing the recommended height of the image
- width: this is a number representing the recommended width of the image
- size: this is a number representing the size of the image in bytes



2.6.2 Operation: getDiagnosisConnectorLocationImage

This operation returns the EOBD connector location image for a specified car.

Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:getDiagnosisConnectorLocationImage>
         <web:carTypeId>26650</web:carTypeId>
      </web:getDiagnosisConnectorLocationImage>
   </soapenv:Body>
</soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getDiagnosisConnectorLocationImageResponse
```

```
xmlns="http://webservice.elecdata.vivid.nl">
        <getDiagnosisConnectorLocationImageReturn>
           <image>
              <height>494</height>
              <imagePath>http://www.workshopdata.com/workshop/images/volkswagen golf iv &amp; bora_dba_romania.svgz</imagePath>
              <size>9536</size>
              <width>593</width>
           </image>
        </getDiagnosisConnectorLocationImageReturn>
     </getDiagnosisConnectorLocationImageResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

2.7 Location

2.7.1 Structure – system

```
ExtLocationSystemArrayContainerV2 {
      ExtSymbolInstanceArrayContainer[]    symbolInstanceArrayContainer;
}
```

- **symbolInstanceArrayContainer**: this is an array containing all linked symbol instances location systems for the selected vesa component;

```
ExtSymbolInstanceArrayContainer {
      SymbolInstanceLocationSystem[]       symbolInstances;
}
```

- **symbolInstances**: this is an array of one or 2 elements representing the symbol instances that a location system might have;

```
SymbolInstanceLocationSystem {
      Integer                              symbol_instance_id;
      String                               symbol_instance_description;
      ExtLocationSystem                    locationSystem;
}
```

- **symbol_instance_id**: this is a number representing the id of the symbol instance component;
- **symbol_instance_description**: this is a string representing the symbol instance component name;
- **locationSystem**: this is the location system containing the symbol instance component with the id specified above and that component should be highlighted in the location schematics linked;

```
ExtLocationSystem {
      Integer            order;
      Integer            id;
      String             description;
      Integer            side;
      LocationImage      systemPositionImage;
```

```
        LocationImage        systemDetailImage;
        String               searchedCompName;
        ExtLocationItem[]    items;
        String               locationId;
        String[]             otherSystemLocationIds;
        String[]             remarks;
}
```

- **order**: this is a number representing the sort order of the ExtLocationSystem objects
- **id**: this is a number that uniquely identifies an ExtLocationSystem object
- **description**: this is a text containing the description of the location (e.g.: “Top view”)
- **side**: this property is a number that indicates the side of the wheel:
    - 0 – both
    - 1 – left
    - 2 – right
    - -1 – unknown
- **systemPositionImage**: this property is a string that contains an URL to a picture that shows the location of the system on the car
- **systemDetailImage**: this property is a string that contains an URL to a picture that shows the detail view of the component location
- **searchedCompName**: this is the code of the component which should for the component that should be highlighted
- **items**: this is a list of complex objects representing the objects in the location scheme
- **locationId**: this is a text that can contain a unique identifier of the location system if necessary
- **otherSystemLocationIds**: when more systems are part of the same group, this property contains the locationIds of the other systems in the same group (for example “sys-9” - “Knock sensor” is part of “Engine management” which contains other systems, like “sys-5” - “Oil level and temperature sensor”)
- **remarks**: this is a list of text objects containing remarks about the location system (e.g.: “Left-hand drive is shown.”)

```
LocationImage {
        String      imagePath;
        Integer     height;
        Integer     width;
        long        size;
}
```

- **imagePath**: this is a text representing the URL of the location picture
- **height**: this is a number representing the height of the picture
- **width**: this is a number representing the width of the picture
- **size**: this is a number representing the size of the image in bytes

```
ExtLocationItem {
        String      description;
        String      type;
        String      location;
```

```
        String       internalLocation;
        String       remark;
        double       value;
        int          sysValue;
        int          id;
}
```

- **description**: this is a text containing the description of a component that appears in the image (for example: “Knock sensor” or “Throttle control motor with position sensor.”)
- **type**: this represents the type of the component and it is used together with the “location” value to create a text that represents an identifier of the component in the the image (for example: “type” can be “fus” and the “location” can be “2”; the identifier then becomes “fus-2” which is the id of the svg element which surrounds the component in the image)
- **location**: this is a text that represents an identifier of the component location; see the “type” property for more details (e.g.: “1”,”3”,”14a”)
- **internalLocation**: not used (it has the same value as “location”)
- **remark**: this is a text that, if not null, gives extra details about the location (e.g.: “In the centre console.”)
- **value**: this is a number that represents the value component of the location; (e.g.: for fuses, it represents the value in Amps; for other types, the value is 0.0)
- **sysValue**: not used (always -1)
- **id**: this is a number that represents an identifier of the location

2.7.2 Structure – system by subject

```
ExtLocationSubjectsArrayContainer {
        ExtLocationSubject[]     locationSubjects;
}

ExtLocationSubject {
        String                       description;
        Integer                      subject;
        ExtLocationSystemGeneric[]   locationSystems;
}
```

- **description**: the description of the location subject (e.g.: “(E)OBD Connector”, “Fuses & Relays”)
- **subject**: this is a number representing the subject:
    - 0 – none
    - 1 – motor management
    - 2 – control module
    - 3 – ground points
    - 4 – EOBD
    - 5 – fuses and relays

- **locationSystems**: this is a list of complex objects containing the location systems

```
ExtLocationSystemGeneric {
      Integer             order;
      Integer             id;
      String              description;
      Integer             side;
      LocationImage       systemPositionImage;
      LocationImage       systemDetailImage;
      ExtLocationItem[]   items;
      String[]            remarks;
      String[]            otherSystemLocationIds;
      String              locationId;
}
```

- **order**: this is a number representing the sort order of ExtLocationSystemGeneric object for a certain subject
- **id**: this is a number representing an identifier for the ExtLocationSystemGeneric objects
- **description**: this is the description of the location system (e.g.:"Top view", "Fuse box in passenger compartment, from MY 1998")
- **side**: this property is a number that indicates the side of the wheel:
  - 0 – both
  - 1 – left
  - 2 – right
  - -1 – unknown
- **systemPositionImage**: this property is a string that contains an URL to a picture that shows the location of the system on the car
- **systemDetailImage**: this property is a string that contains an URL to a picture that shows the detail view of the component location
- **items**: this is a list of complex objects representing the objects in the location scheme
- **remarks**: this is a list of text objects containing remarks about the location system (e.g.: "Left-hand drive is shown.")
- **otherSystemLocationIds**: when more systems are part of the same group, this property contains the locationIds of the other systems in the same group (for example "sys-9" - "Knock sensor" is part of "Engine management" which contains other systems, like "sys-5" - "Oil level and temperature sensor")
- **locationId**: this is a text that can contain a unique identifier of the location system if necessary

Note! The **ExtLocationSystemGeneric** objects have a very similar structure with **ExtLocationSystem**

2.7.3 Operation: getLocationSystemV2

This operation returns an array of symbol instances location systems for a specified car and component. Each symbol instance location system can have one or two possible symbol instances (symbol instance and symbol instance 2).

Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:data="http://data.webservice.workshop.vivid.nl">
<soapenv:Header/>
      <soapenv:Body>
        <data:getLocationSystemV2>
          <data:vrid>[vrid]</data:vrid>
          <data:carTypeId>102000423</data:carTypeId>
          <data:elecDataCompId>301247936</data:elecDataCompId>
          <data:language>en</data:language>
       </data:getLocationSystemV2>
   </soapenv:Body>
   </soapenv:Envelope>
```

Note! The “elecDataCompId” is the symbol instance id (ComponentWs.si or Component.symbolInstanceId)

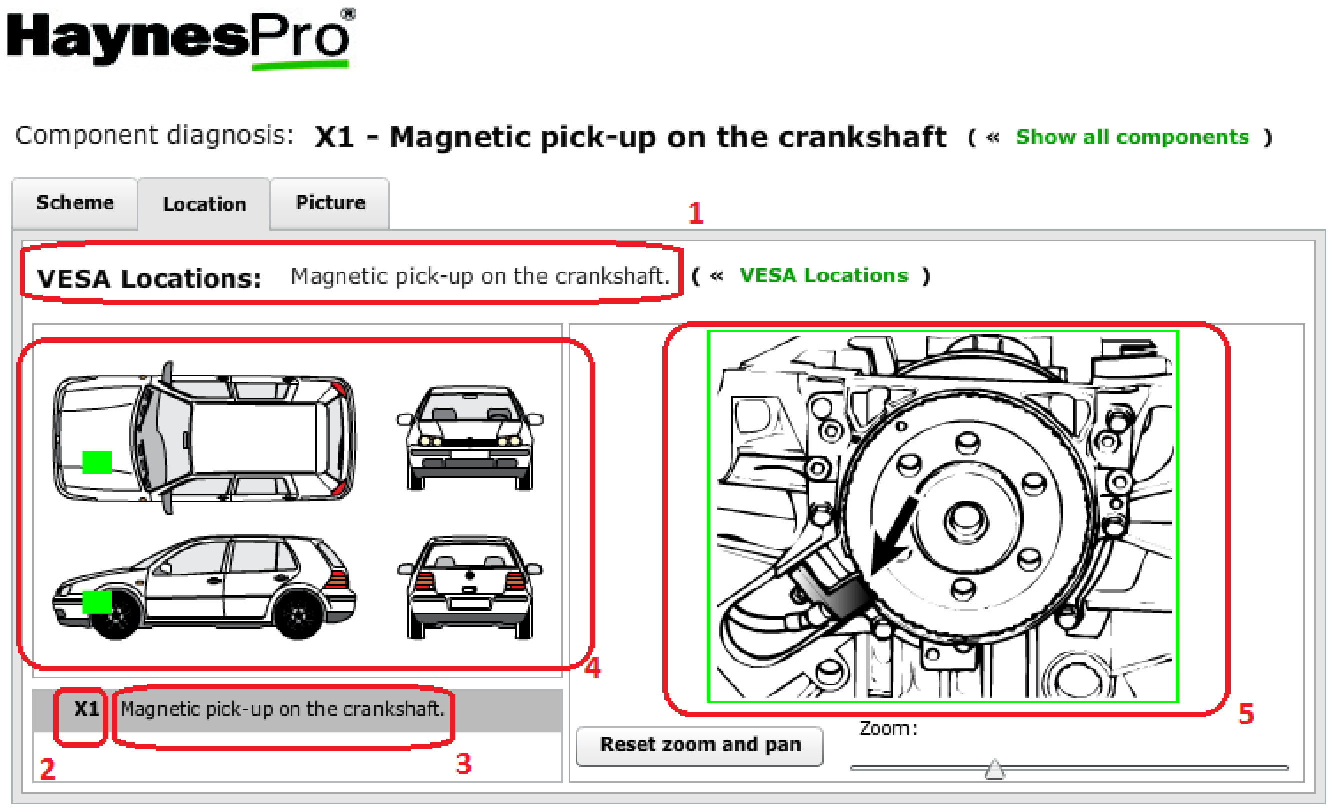
- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getLocationSystemV2Response
xmlns="http://data.webservice.workshop.vivid.nl">
         <getLocationSystemV2Return>
            <status xsi:nil="true"/>
            <symbolInstanceArrayContainer>
               <symbolInstanceArrayContainer>
                  <status xsi:nil="true"/>
                  <symbolInstances>
                        <symbolInstances>
                           <locationSystem>
                              <description>Fuse and relay box in engine
compartment, (2009 - ), LHD (Left-hand drive view)</description>
                              <id>301003939</id>
                              <items>
                             <items>
                                    <description>Fuel pump</description>
                                    <id>301051857</id>
                                    <internalLocation>21</internalLocation>
                                    <location>21</location>
                                    <remark/>
                                    <sortOrder>0</sortOrder>
                                    <sysValue>-1</sysValue>
                                    <type>fus</type>
                                    <value>20.0</value>
```

```
                                </items>
                                  …
                            </items>
      <locationId xsi:nil="true"/>
                             <order>2</order>
                             <otherSystemLocationIds xsi:nil="true"/>
                             <remarks xsi:nil="true"/>
                             <searchedCompName>43</searchedCompName>
                             <side>1</side>
                             <systemDetailImage>
                                <height>432</height>
  <imagePath>c:\workshop\images\301010543.svgz</imagePath>
                                <size>1984</size>
                                <width>801</width>
                             </systemDetailImage>
                             <systemPositionImage>
                                <height>287</height>
  <imagePath>c:\workshop\images\301010471.svgz</imagePath>
                                <size>5830</size>
                                <width>643</width>
                             </systemPositionImage>
                          </locationSystem>
                          <locationSystems xsi:nil="true"/>
                          <symbol_instance_description>07 Fuse F43 PDC (2009 - )</symbol_instance_description>
                          <symbol_instance_id>301247936</symbol_instance_id>
                       </symbolInsances>
                    </symbolInsances>
                 </symbolInstanceArrayContainer>
                 <symbolInstanceArrayContainer>
                    <status xsi:nil="true"/>
                    <symbolInsances>
                       <symbolInsances>
                          <locationSystem>
                             <description>Fuse and relay box in engine compartment, (2007 - 2008), LHD (Left-hand drive view)</description>
                             <id>301003926</id>
                             <items>
                              …
                             </items>
                             <locationId xsi:nil="true"/>
                             <order>3</order>
                             <otherSystemLocationIds xsi:nil="true"/>
                             <remarks xsi:nil="true"/>
                             <searchedCompName>16</searchedCompName>
                             <side>1</side>
                             <systemDetailImage>
                                <height>321</height>
  <imagePath>c:\workshop\images\301010475.svgz</imagePath>
                                <size>3140</size>
```

```xml
                                   <width>701</width>
                                </systemDetailImage>
                                <systemPositionImage>
                                   <height>287</height>
  <imagePath>c:\workshop\images\301010471.svgz</imagePath>
                                   <size>5830</size>
                                   <width>643</width>
                                </systemPositionImage>
                             </locationSystem>
                             <locationSystems xsi:nil="true"/>
                             <symbol_instance_description>07 Fuse F16
  PDC</symbol_instance_description>
                             <symbol_instance_id>301247936</symbol_instance_id>
                          </symbolInsances>
                       </symbolInsances>
                    </symbolInstanceArrayContainer>
                 </symbolInstanceArrayContainer>
              </getLocationSystemV2Return>
           </getLocationSystemV2Response>
        </soapenv:Body>
     </soapenv:Envelope>
```

The difference between V1 and V2 of this operation is that in the last one version instead of an ExtLocationSystemArrayContainer object an ExtSymbolInstanceArrayContainer object is returned (See the structure of the Location – system for more details). This means that for each location system linked to the selected component and type we can have one or two symbol instances location systems(when we click on the first symbol instance the component with the specified symbol instance description is highlighted, the same happens when we click on the symbol instance 2 if this exists).



1 - <description>Magnetic pick-up on the crankshaft.</description> (line 7)
2 - <location>X1</location> (line 14)
3 - <description>Magnetic pick-up on the crankshaft.</description> (line 11)
4 - <systemPositionImage> (line 37)
5 - <systemDetailImage> (line 31)

- fragment from “systemDetailImage”:

```
...
<g id="mmp-X1">
                <path i:knockout="Off" fill="none" stroke="#F3160D"
d="M0.5,317.5h401V0.5H0.5V317.5z"/>
          </g>
…
```

- fragment from “systemPositionImage”:

```
...
<g id="sys-6">
    <rect x="39.454" y="100.059" fill="none" stroke="#CA2034" stroke-
miterlimit="2.4142" width="18.265" height="11.939"/>
    <rect x="39.454" y="242.979" fill="none" stroke="#CA2034" stroke-
miterlimit="2.4142" width="18.265" height="11.939"/>
</g>
…
```

2.7.4 Operation: getLocationSystemBySubject

This operation returns all the location systems for a specified group on a combination of subjects.

Example:

- request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:web="http://webservice.elecdata.vivid.nl">
   <soapenv:Header/>
   <soapenv:Body>
      <web:getLocationSystemBySubject>
         <web:carTypeId>26650</web:carTypeId>
         <web:showMMP>false</web:showMMP>
         <web:showFusesAndRelays>true</web:showFusesAndRelays>
         <web:showControlUnit>false</web:showControlUnit>
         <web:showGroundingPoints>false</web:showGroundingPoints>
            <web:language>en</web:language>
         </web:getLocationSystemBySubject>
      </soapenv:Body>
   </soapenv:Envelope>
```

- response:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <soapenv:Body>
      <getLocationSystemBySubjectResponse
xmlns="http://webservice.elecdata.vivid.nl">
         <getLocationSystemBySubjectReturn>
            <locationSubjects>
               <item>
                  <description>Fuses and Relays</description>
                  <locationSystems>
                     <item>
                            <description>Fuse box in passenger compartment, from
MY 1998</description>
                            <id>175</id>
                            <items>
                               <item>
                                  <description>Heated washers. Heated
mirror(s).</description>
                                  <id>2188</id>
                                  <internalLocation>1</internalLocation>
                                  <location>1</location>
                                  <remark/>
                                  <sortOrder>20</sortOrder>
                                  <sysValue>-1</sysValue>
                                  <type>fus</type>
                                  <value>10.0</value>
```

```
                              </item>
                              <item>
                                 <description>Direction indicators.</description>
                                 <id>21835</id>
                                 <internalLocation>2</internalLocation>
                                 <location>2</location>
                                 <remark/>
                                 <sortOrder>30</sortOrder>
                                 <sysValue>-1</sysValue>
                                 <type>fus</type>
                                 <value>10.0</value>
                              </item>
                              ...
                           </items>
                           <locationId xsi:nil="true"/>
                           <order>0</order>
                           <otherSystemLocationIds xsi:nil="true"/>
                           <remarks>
                              <item>Left-hand drive is shown.</item>
                           </remarks>
                           <side>0</side>
                           <systemDetailImage>
                              <height>248</height>
                              <imagePath>http://www.workshopdata.com/workshop/images/fuse_golf4_98+_1_dba_uk[Converted].svgz</imagePath>
                              <size>4176</size>
                              <width>428</width>
                           </systemDetailImage>
                           <systemPositionImage>
                              <height>494</height>
                              <imagePath>http://www.workshopdata.com/workshop/images/loc_dashbleft_side_dba_ellemeet.svgz</imagePath>
                              <size>8331</size>
                              <width>593</width>
                           </systemPositionImage>
                        </item>
                        ...
                     </locationSystems>
                     <subject>5</subject>
                  </item>
               </locationSubjects>
               <status xsi:nil="true"/>
            </getLocationSystemBySubjectReturn>
         </getLocationSystemBySubjectResponse>
      </soapenv:Body>
   </soapenv:Envelope>
```

HaynesPro

omponents | Fault code diag... | Wiring Diagrams | Repair Manuals | **VESA Locations**

Control Modules

Fuses and Relays

Fuse box in passenger compartment, from 1998

Fuse box in engine compartment, from MY 1998

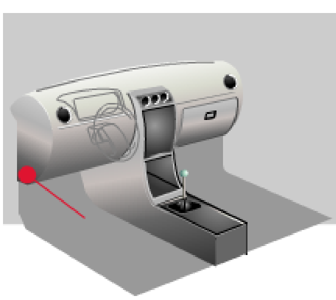
Fuse and relay box in passenger compartment, from MY 1998 (Left hand drive view)

Fuse and relay box in passenger compartment, from MY 1998 (Right hand drive view)

(E)OBD Connector

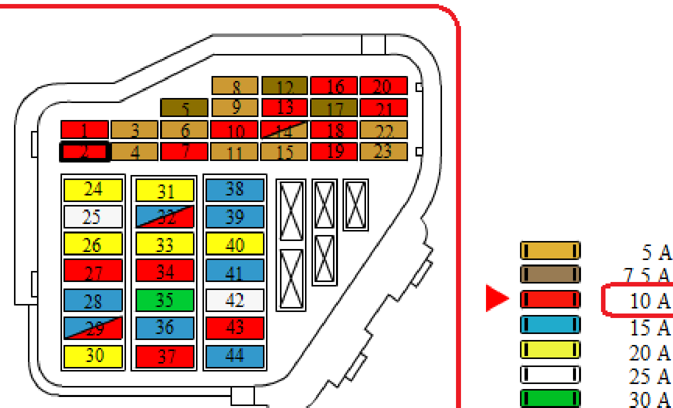
Engine Management

VESA Locations: Fuse box in passenger compartment, from 1998 (« **Change location system**)



Left-hand drive is shown.

1	Heated washers. Heated mirror(s).
2	Direction indicators.
3	Lighting.
4	Number plate light(s).
5	Comfort system electrics.



Reset zoom and pan

Zoom: